



GRADIENT PROJECTION METHODS FOR CONTINUAL LEARNING IN HYPERNETWORKS

OVERCOMING CHUNK-INDUCED GRADIENT
PATHOLOGIES VIA PROJECTION
NORMALIZATION, AND THE INTRODUCTION OF
IFOPNG

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Abstract

This thesis addresses the fundamental incompatibility of applying gradient projection-based continual learning (CL) methods to hypernetwork architectures, establishing a novel framework for stable subspace optimization. Traditional projection methods collapse due to a structural-numerical pathology where the architectural necessity of weight-chunking, required to avoid an output parameter explosion, forces the model to be produced in hundreds or thousands of sequential chunks before a single projection step can execute. This repetitive chunk-generation loop accumulates an exceptionally large gradient volume within the memory matrix, blowing up the condition number of the projection correlation matrix beyond 10^{10} , causing catastrophic inversion failures that the standard regularization parameter λ cannot rescue.

The primary research objective is to eliminate these numerical singularities and systematically evaluate whether combining hard geometric projections on top of task-conditioned hypernetworks and their output-scoped functional regularizers yields a constructive synergy. To achieve this, we introduce *projection-scoped normalization*: gradient columns stored in the memory matrix G are rescaled to unit ℓ_2 norm, and the diagonal empirical Fisher vector is rescaled by its absolute maximum, together stabilizing the metric tensor within the local projection algebra. We also introduce iFOPNG, an inertial variant of Fisher-Orthogonal Projected Natural Gradient Descent (Garg, Kolhe, Peng, & Gopalam, 2026) that replaces the current-task Fisher F_{new} with the combined curvature $F_c = F_{\text{new}} + F_{\text{old}}$, inspired by the parameter-inertia concept of Elastic Weight Consolidation. Methods are benchmarked on sequential Split and Permuted MNIST, Split-CIFAR10 and Split-CIFAR100. The empirical findings reveal that projection normalization completely resolves matrix inversion collapse, and that iFOPNG consistently outperforms FOPNG across all

benchmarks; in the standard-capacity hypernetwork setting only iFOPNG surpasses plain SGD, while most other projection variants fall below it. A standalone hypernetwork optimized with Adam and Oswald, Henning, Grewe, and Sacramento’s functional regularizer consistently surpasses all projection variants, indicating that momentum, which projection methods forgo by design, is the decisive factor in compressed generative weight spaces.

1 DATA SOURCE, ETHICS, CODE, AND TECHNOLOGY STATEMENT

Data Source. The MNIST (owned by LeCun et al.) and CIFAR (CIFAR-10 and CIFAR-100, owned by Krizhevsky et al.) datasets are public, anonymized datasets licensed for research. These datasets form the basis for the Split-MNIST, Permuted MNIST, Split-CIFAR10, and Split-CIFAR100 benchmarks evaluated in this work. No human or animal data was collected. Data was acquired via the `torchvision.datasets` API.

FIGURES. All figures and plots were created entirely by the author using `matplotlib` and `Weights & Biases`.

CODE. The baseline FOPNG framework was adapted from Garg et al. (2026); all other execution infrastructure was implemented by the author (<https://github.com/kelares/HyperFisher>). Libraries used: Python (v3.10), PyTorch (v2.x), Torchvision, Weights & Biases, NumPy, Matplotlib, and Tqdm.

TECHNOLOGY. The manuscript was typeset in $\text{\LaTeX}$ using BibTeX. During the preparation of this work, the author(s) used Gemini and Claude in order to optimize academic phrasing, outline structure, paraphrase transitions, and check grammar. After using these tools/services, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the thesis. AI Citations: Google. (2026). *Gemini* [Large language model]. <https://gemini.google.com>; Anthropic. (2026). *Claude* [Large language model]. <https://claude.ai>.

2 INTRODUCTION

The capacity for continuous, sequential adaptation without the eradication of historical knowledge remains an architectural cornerstone of biological intelligence, yet it presents a fundamental barrier for artificial neural networks. In deep learning paradigms, sequential training across non-stationary data distributions triggers a systemic pathology known as *catastrophic*

trophic forgetting (McCloskey & Cohen, 1989). When a network optimizes its parameters for an incoming task, gradient updates blindly overwrite the weight configurations essential for retaining prior task spaces, shifting the internal representation manifolds away from historical objective minima.

Beyond its theoretical interest, this limitation carries direct practical cost. Deployed systems must either be retrained from scratch whenever the data distribution shifts — at substantial computational and energetic expense — or retain growing archives of historical data for replay, an option foreclosed in privacy-regulated and on-device settings where past data cannot be stored. Continual learning methods that preserve prior competence without replay therefore address both the economic burden of repeated retraining and the deployment of adaptive models under data-retention constraints.

Modern continual learning (CL) literature is conventionally structured into three core algorithmic families: replay-based strategies that leverage episodic data buffers, regularization-based frameworks that penalize parameter deviations, and architectural or parameter-isolation methods (De Lange et al., 2022). Within the meta-learning domain, hypernetworks (Ha, Dai, & Le, 2017) have emerged as an elegant paradigm. Hypernetworks circumvent parameter-space conflicts by utilizing a centralized, lightweight meta-model to generate the target network’s complete parameter weights conditioned on task embeddings, as formalized by Oswald et al. (2020). Concurrently, within the field of geometric optimization, gradient projection methods, exemplified by the recently proposed Fisher-Orthogonal Projected Natural Gradient Descent (FOPNG) (Garg et al., 2026), explicitly constrain subsequent updates to lie orthogonal to the subspace of parameter directions critical for previous tasks, with the geometry of orthogonality ranging from Euclidean (OGD) to Fisher-Riemannian (FOPNG).

Despite the mathematical elegance of both hypernetworks and gradient projection methods, their structural intersection uncovers a critical and previously unexamined optimization pathology. Generating a target network’s full weight vector in a single forward pass would require an output layer whose width equals the total number of target network parameters, which is computationally prohibitive for any non-trivial architecture. Chunked hypernetworks therefore partition generation into K sequential sub-passes, each emitting a fixed-size segment of the weight vector from the same shared generator; because every sub-pass executes before any gradient update, their contributions accumulate additively in the shared meta-model parameters, inflating the gradient magnitude proportionally to K . This gradient inflation propagates into the projection algebra: both the gradient columns stored in memory matrix G and the diagonal Fisher estimate

$\hat{F}$ grow in scale with K , rendering the regularization term λI negligible relative to the inflated projection matrix $G^\top \hat{F} G$. The condition number of this matrix exceeds 10^{10} , causing catastrophic inversion failures.

Research Objectives and Algorithmic Framework

The primary research objective of this thesis is to systematically resolve this linear-algebraic collapse, map the initialization mechanics of meta-projection pipelines, and evaluate whether the integration of hard geometric constraints on top of soft functional regularizers yields an optimized sequential learning system.

To achieve this, we introduce *projection-scoped normalization* (column-wise ℓ_2 orthonormalization of G combined with absolute-maximum scaling of $\hat{F}$) and iFOPNG, an inertial FOPNG variant using the combined metric $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$. These contributions enable stable gradient projection in chunked hypernetworks, with direct applicability to on-device sequential learning where retaining adaptive momentum is critical.

Research Questions

To systematically navigate these boundaries, this thesis addresses the following primary research question:

What numerical pathologies arise when gradient projection methods are applied to chunked hypernetwork architectures, and do modifications to the Fisher metric construction—specifically an inertial combined metric and a monotonic accumulation operator—yield principled improvements over standard projection baselines in continual learning?

This is decomposed into four sub-research questions (Sub-RQs):

- Sub-RQ1 (Synergy vs. Conflict):** Does the joint integration of hard gradient projection with the hypernetwork’s soft functional regularizer yield a constructive synergy, or does a standalone regularized hypernetwork offer a more efficient optimization frontier?
- Sub-RQ2 (Gradient Pathology):** How does the architectural necessity of spawning K sequential weight-chunks inflate the gradient and Fisher magnitudes accumulated before a single projection step, and does rescaling both components independently within the projection algebra suffice to prevent matrix inversion collapse?

- Sub-RQ3 (Parameter Inertia):** Does replacing FOPNG’s task-specific metric $\hat{F}_{\text{new}}$ with the combined inertial metric $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$ yield consistent retention improvements across benchmarks?
- Sub-RQ4 (Fisher Accumulation):** Does accumulating $\hat{F}_{\text{old}}$ via a non-decaying element-wise maximum operator maintain long-horizon retention more effectively than an exponential moving average, and does the monotonic growth of the maximum operator impose a measurable plasticity cost over extended task sequences?

Summary of Empirical Findings

Projection-scoped normalization fully resolves the matrix inversion pathology, recovering stable projection performance across all seeds. The inertial combined metric $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$ consistently improves retention over FOPNG, with the advantage scaling monotonically with inter-task Fisher overlap. The MAX accumulation operator outperforms EMA on long task sequences and requires no additional hyperparameters. In the standard-capacity hypernetwork setting, iFOPNG surpasses plain SGD but falls short of Adam; under extreme compression the gap to Adam widens to 20–40 pp and most projection methods drop below SGD entirely.

3 RELATED WORK

This section builds the theoretical and empirical context for the thesis, tracing a path from foundational continual learning challenges through regularization and projection methods, to hypernetwork architectures and the structural gap at their intersection.

3.1 *Continual Learning and the Stability-Plasticity Dilemma*

Modern machine learning systems are predominantly trained on fixed datasets drawn from a single stationary distribution — a condition that rarely holds in deployment, where agents must adapt to non-stationary data streams while retaining prior knowledge. The fundamental tension between these two objectives is formalized as the *stability-plasticity dilemma* (Grossberg, 1980): an agent must remain plastic enough to absorb novel distributions yet stable enough to resist the erasure of established representations. When standard architectures trained by empirical risk minimization are exposed to sequential data, they suffer *catastrophic forgetting* (McCloskey & Cohen, 1989) — abrupt and near-total loss of previously learned capabilities upon exposure to new task data.

The continual learning literature addresses this across three canonical scenarios (De Lange et al., 2022): *task-incremental learning*, where the task identity is available at both training and test time; *class-incremental learning*, where task identity is withheld at inference; and *domain-incremental learning*, where the output space is shared but the input distribution shifts continuously. This thesis focuses on the task-incremental setting, the scenario in which projection-based methods and hypernetwork architectures have been most thoroughly evaluated and in which structural comparisons between them are most tractable.

3.2 Evaluation Protocols: Multi-Head and Single-Head Designs

Within the task-incremental setting, two evaluation protocols govern how the network’s output layer is organized. In the *multi-head* protocol, each task is assigned a dedicated output head selected at test time using the known task identity, eliminating output-level interference between tasks. In the *single-head* protocol, all tasks share one output layer, requiring the backbone alone to resolve inter-task interference — a substantially harder problem. Farquhar and Gal (2018) demonstrated that multi-head evaluations systematically inflate apparent method performance by removing a major source of forgetting, and argued that single-head evaluation is the more rigorous protocol.

This thesis uses multi-head benchmarks for all Split-CIFAR experiments to match the evaluation conditions of Garg et al. (2026) and enable direct numerical comparison. For Split-MNIST, both protocols are evaluated: the multi-head variant replicates Garg et al.’s conditions, while the single-head variant follows Farquhar and Gal (2018)’s recommendation. We replicate Garg et al.’s task-incremental single-head protocol — a shared two-neuron output head with task identity provided to the evaluation harness at test time — subject to a necessary first-task learning rate correction detailed in Section 4.6.1.

3.3 Regularization-Based Approaches

The dominant paradigm for addressing catastrophic forgetting is *parameter regularization*, of which Elastic Weight Consolidation (EWC) (Kirkpatrick et al., 2017) is the canonical representative. EWC estimates the structural importance of each weight using the diagonal of the empirical Fisher

Information Matrix $\hat{F}$, and applies a soft quadratic penalty that resists changes to parameters important to prior tasks:

$$\mathcal{L}_{\text{EWC}}(\theta) = \mathcal{L}_t(\theta) + \frac{\lambda}{2} \sum_i \hat{F}_i (\theta_i - \theta_{i,t-1}^*)^2. \quad (1)$$

EWC is computationally efficient and broadly applicable, and serves as the standard reference point against which new continual learning methods are measured. Its principal limitation is that soft penalties permit bounded parameter drift: cumulative updates across a long task sequence gradually erode the protection of earlier task spaces, a failure mode that becomes increasingly pronounced as T grows (De Lange et al., 2022).

3.4 Gradient Projection Methods

To enforce strict, non-negotiable protection for prior knowledge rather than relying on penalties that can be overcome by sufficiently large gradients, a second paradigm uses explicit *gradient projection*. Orthogonal Gradient Descent (OGD) (Farajtabar, Azizan, Mott, & Li, 2020) maintains a memory matrix $G = [g_1 \mid \cdots \mid g_K]$ of historical task gradient directions and projects incoming gradients onto the orthogonal complement of $\text{Col}(G)$ before each update:

$$\tilde{g} = g - G(G^\top G)^{-1}G^\top g. \quad (2)$$

This provides a hard guarantee: the update cannot increase the loss on previously seen tasks. Generalization bounds established within the Neural Tangent Kernel regime confirm that OGD provides exact protection in linear and mildly nonlinear settings (Bennani, Doan, & Sugiyama, 2020).

The mathematical appeal of projection methods over EWC lies precisely in this hardness: where EWC applies a soft penalty that can be overwhelmed, projection methods impose a geometric constraint that is either satisfied or not. However, standard Euclidean projection treats the parameter space as flat and assigns equal importance to all directions. Fisher-Orthogonal Projected Natural Gradient Descent (FOPNG) (Garg et al., 2026) addresses this by performing the projection in the Riemannian geometry defined by the Fisher metric, so that the orthogonality constraint respects the curvature of the loss surface. The related Fisher Natural Gradient (FNG) (Garg et al., 2026) applies the natural gradient without projection, and Orthogonal Natural Gradient (ONG) (Yadav, Mendoza, & Korrapati, 2025) combines natural gradient preconditioning with Euclidean projection — sitting structurally between OGD and FOPNG.

3.5 Hypernetworks for Continual Learning

An orthogonal architectural response to catastrophic forgetting replaces the multi-head output design entirely with a *hypernetwork* (Ha et al., 2017): a meta-model $\mathcal{H}(\cdot; \phi)$ trained to synthesize the complete weight vector $\theta_t = \mathcal{H}(e_t; \phi)$ of a target network, conditioned on a task embedding e_t . Each task receives a unique weight configuration generated on demand, rather than a dedicated output head, achieving implicit task separation through the generative bottleneck and impressive parameter compression ratios relative to storing separate networks per task. Generating the full θ_t in a single forward pass is computationally intractable for large target networks, so Oswald et al. (2020) introduced *weight chunking*: the generator iterates over a secondary chunk embedding c_k , producing C_s target parameters per pass and assembling the full weight vector by concatenation. To prevent the hypernetwork from forgetting how to generate prior task configurations, a soft *functional regularizer* is applied in the output space of the generator:

$$\mathcal{L}_{\text{reg}} = \beta \sum_{s=1}^{t-1} \|\mathcal{H}(\phi, e_s) - \theta_s^*\|_2^2, \quad (3)$$

where θ_s^* denotes the weight snapshot saved after task s . This penalty can be interpreted as EWC applied at the meta level: rather than anchoring individual network parameters, it anchors the generated weight configurations for each previously seen task, penalizing drift in the space where task-relevant quantities actually live. Recent extensions have refined the hypernetwork approach: Partial Hypernetworks (Hemati, Lomonaco, Bacciu, & Borth, 2023) reduce the computational overhead of full weight generation, and Prototype Augmented Hypernetworks (Fuente et al., 2025) augment task embeddings with prototype representations to further stabilize sequential learning.

3.6 Identified Gaps and Motivating Pathologies

Whether hard geometric projection constraints are compatible with the soft functional regularization already employed by hypernetworks, or whether layering both simultaneously over-constrains the generator’s plasticity, has not been investigated (Sub-RQ1).

The architectural necessity of weight-chunking introduces a gradient accumulation pathology — inflated condition numbers in the projection kernel — that is unique to the hypernetwork and projection methods combination and has no prior treatment in the literature (Sub-RQ2).

Existing projection methods enforce a rigid subspace boundary on historical gradient directions; whether replacing this boundary with a

combined Fisher metric encoding both old and new task curvature yields consistent retention improvements across benchmarks has not been examined (Sub-RQ3).

The historical Fisher estimate underlying Fisher-based projection depends on a choice of accumulation operator; whether a monotonic maximum or a decaying exponential moving average better preserves long-horizon retention has not been directly compared (Sub-RQ4).

These four gaps jointly motivate the sub-research questions stated in Section 2 and the experimental programme described in Section 4.6.

4 METHODS

Conceptual motivation and positioning relative to prior work are covered in Section 3; we restrict our attention here to the exact algorithmic instantiations evaluated.

4.1 Continual Learning Protocol

4.1.1 Task Setup and Evaluation Metrics

A continual learning agent receives a sequence of T tasks $\mathcal{T}_1, \mathcal{T}_2, \dots, \mathcal{T}_T$ presented in order. At task t , only the data from $\mathcal{T}_t$ is available; prior task data is not retained. After each task is trained, the model is evaluated on all tasks seen so far. We report two aggregate metrics: *average accuracy*

$$A_t = \frac{1}{t} \sum_{i=1}^t a_{t,i}, \quad (4)$$

and *backward transfer*

$$\text{BWT} = \frac{1}{T-1} \sum_{t=2}^T (a_{T,t} - a_{t,t}), \quad (5)$$

where $a_{i,j}$ denotes accuracy on task j immediately after training task i . Negative BWT quantifies catastrophic forgetting; a value of zero indicates perfect retention.

Reporting both metrics jointly is necessary because each can mask failure modes invisible to the other: a method that forecloses new learning to preserve old tasks achieves near-zero BWT alongside collapsed accuracy, while a method that overfits each new task sustains competitive accuracy at the cost of large negative BWT on earlier ones.

4.1.2 Multi-Head and Single-Head Evaluation

Following Farquhar and Gal (2018), who demonstrated that multi-head evaluations can inflate apparent method performance, we report results under both protocols for MNIST-based benchmarks.

4.2 Fisher Information Matrix and Regularization-Based Continual Learning

4.2.1 Fisher Information Matrix

All Fisher-based methods in this thesis use the *diagonal empirical approximation*

$$\hat{F}_i \approx \frac{1}{N} \sum_{n=1}^N \left(\frac{\partial \mathcal{L}^{(n)}}{\partial \theta_i} \right)^2, \quad (6)$$

estimated over N held-out samples. The full FIM is intractable for our network sizes; the diagonal approximation is standard across all evaluated methods (Garg et al., 2026; Kirkpatrick et al., 2017).

4.2.2 Elastic Weight Consolidation

EWC (Kirkpatrick et al., 2017) augments the task loss with a Fisher-weighted quadratic penalty:

$$\mathcal{L}_{\text{EWC}}(\theta) = \mathcal{L}_t(\theta) + \frac{\lambda}{2} \sum_i \hat{F}_i (\theta_i - \theta_{i,t-1}^*)^2, \quad (7)$$

where θ_{t-1}^* denotes the parameter snapshot after task $t-1$. EWC operates exclusively through parameter-space penalties and maintains no gradient subspace; it is included as a regularization baseline employing a different mechanism from projection. We use $\lambda = 10^{-3}$ for all hypernetwork configurations and reproduce the per-benchmark values from Garg et al. (2026) Table 1 for standalone evaluations. Fisher estimates are computed over the full training set for MNIST-based benchmarks and over $N=1024$ samples for CIFAR benchmarks.

4.3 Gradient Projection Methods

This subsection presents the full suite of gradient-based methods evaluated in this thesis. SGD and FNG establish unconstrained lower bounds with no forgetting-protection mechanism; OGD through iFOPNG form the projection family that is the primary focus of this work.

4.3.1 Stochastic Gradient Descent

SGD applies vanilla gradient descent with no forgetting protection:

$$\Delta\theta_{\text{SGD}} = -\eta \nabla_{\theta} \mathcal{L}_t. \quad (8)$$

Adam (Kingma & Ba, 2015) is included as an adaptive-rate variant.

4.3.2 Natural Gradient and Fisher Natural Gradient

Natural gradient descent (Amari, 1998) preconditions the gradient with the inverse FIM, respecting the curvature of the statistical manifold:

$$\Delta\theta_{\text{nat}} = -\eta \hat{F}^{-1} \nabla_{\theta} \mathcal{L}. \quad (9)$$

FNG (Garg et al., 2026) applies a trust-region constraint to this Riemannian step:

$$v_{\text{FNG}}^* = -\eta \frac{\hat{F}^{-1} g}{\sqrt{g^{\top} \hat{F}^{-1} g}}. \quad (10)$$

FNG imposes no projection subspace and constitutes no continual learning mechanism; it is included to isolate the contribution of Fisher preconditioning from that of gradient projection.

4.3.3 Orthogonal Gradient Descent

OGD (Farajtabar et al., 2020) maintains a memory matrix $G = [g_1 \mid \cdots \mid g_K] \in \mathbb{R}^{p \times K}$ of past task gradient directions and updates parameters via the current gradient projected onto the orthogonal complement of $\text{Col}(G)$:

$$\tilde{g} = g - G(G^{\top} G)^{-1} G^{\top} g, \quad \Delta\theta_{\text{OGD}} = -\eta \tilde{g}. \quad (11)$$

OGD operates in Euclidean space, treating all parameter directions symmetrically.

GRADIENT MEMORY MANAGEMENT. In all projection methods, G is maintained as a fixed-budget basis of at most M columns. Garg et al. (2026) set $M = T \times K$ across all benchmarks, so compression never triggers in their experiments. We address this unexamined regime explicitly: incoming gradient batches are appended directly to G ; when the column count exceeds M , the full basis is recompressed via truncated SVD and $G \leftarrow U_{:,1:M}$, retaining the M directions of greatest variance in the accumulated gradient memory. When projection-scoped normalization is enabled, QR decomposition is additionally applied to each incoming batch at insertion time (Section 4.5.1).

4.3.4 Orthogonal Natural Gradient

ONG (Yadav et al., 2025) applies OGD-style projection to the natural gradient rather than to the raw gradient, combining Fisher preconditioning with a Euclidean orthogonality constraint:

$$\Delta\theta_{\text{ONG}} = -\eta \left(\hat{F}^{-1}g - G(G^\top G)^{-1}G^\top \hat{F}^{-1}g \right). \quad (12)$$

Unlike FOPNG, the projection is measured in Euclidean rather than Fisher metric space.

Following the diagonal Fisher approximation used throughout this thesis, we substitute the EKFac inverse of Yadav et al. (2025) with the diagonal empirical Fisher; results are not directly comparable to the original implementation.

4.3.5 Fisher-Orthogonal Projected Natural Gradient Descent

FOPNG (Garg et al., 2026) executes projection in the Riemannian metric defined by the Fisher:

$$v_{\text{FOPNG}}^* = -\eta \frac{\hat{F}^{-1}Pg}{\sqrt{(Pg)^\top \hat{F}^{-1}Pg}}, \quad P = I - \hat{F}_{\text{old}} G \left(G^\top \hat{F}_{\text{old}} \hat{F}^{-1} \hat{F}_{\text{old}} G + \lambda I \right)^{-1} G^\top \hat{F}_{\text{old}}, \quad (13)$$

where $\hat{F} \equiv \hat{F}_{\text{new}}$ is the current-task Fisher, $\hat{F}_{\text{old}}$ is the Fisher under which the stored directions were collected, and λ is added to the diagonal for numerical stability. This is the pre-Fisher form of the practical algorithm of Garg et al. (2026), in which the stored basis carries its collection-time Fisher weighting; it is the variant implemented and evaluated throughout this thesis, and it makes the relationship to iFOPNG exact: the latter is obtained by the single substitution $\hat{F}_{\text{new}} \rightarrow F_c$ in the preconditioner, the trust region, and the projection kernel. We reproduce the per-benchmark learning rates and λ values from Garg et al. (2026) Table 1 for standalone configurations, and use a uniform $\text{lr}=10^{-3}$, $\lambda=10^{-3}$ for all hypernetwork configurations.

4.3.6 Inertial FOPNG

In FOPNG, the parameter update takes the form

$$v = c \hat{F}_{\text{new}}^{-1}(g - u), \quad (14)$$

where $\hat{F}_{\text{new}}$ serves as the Riemannian metric: both the trust-region constraint $\|v\|_{\hat{F}_{\text{new}}} = \eta$ and the projection objective $\|g - \hat{F}_{\text{new}}^{-1}u\|_{\hat{F}_{\text{new}}}^2$ are measured in the current task’s Fisher geometry. This is natural when no prior tasks exist, but creates an asymmetry once history accumulates: the constraint subspace $\text{Col}(G)$ is built from gradient directions collected under

$\hat{F}_{\text{old}}$, yet the projection metric assigns zero curvature to parameters that were important in previous tasks but uninformative on the current one.

iFOPNG is introduced in this thesis as a variant of FOPNG that resolves this asymmetry by adopting the *combined metric*

$$F_c := \hat{F}_{\text{new}} + \hat{F}_{\text{old}}, \quad (15)$$

where $\hat{F}_{\text{old}}$ is maintained as the element-wise maximum of all per-task Fisher diagonals accumulated so far. Replacing (14) with $v = c F_c^{-1}(g - u)$, the constrained trust-region update becomes

$$v^* = -\eta \frac{F_c^{-1} P g}{\sqrt{(P g)^\top F_c^{-1} P g}}, \quad P = I - \hat{F}_{\text{old}} G A^{-1} G^\top \hat{F}_{\text{old}}, \quad (16)$$

where $A = G^\top \hat{F}_{\text{old}} F_c^{-1} \hat{F}_{\text{old}} G$. A formal derivation is provided in Appendix A.

The combined metric F_c assigns each coordinate i an effective mass $(\hat{F}_{\text{new}})_i + (\hat{F}_{\text{old}})_i$, suppressing the natural-gradient step size in directions important to any prior task. We call this property *parameter inertia*: accumulated Fisher importance makes a coordinate harder to displace without any explicit penalty term. In direct contrast to EWC (Kirkpatrick et al., 2017), which pulls parameters toward a stored reference point θ^* , iFOPNG maintains no reference point: a parameter that mattered before is not pulled back, it is simply made harder to displace.

4.4 Hypernetwork Architecture

4.4.1 Chunked Hypernetworks

Generating θ_t in a single forward pass would require an output layer of dimension $|\theta_t|$, which is computationally prohibitive for any non-trivial target architecture. *Weight chunking* (Oswald et al., 2020) resolves this by iterating over a secondary chunk embedding c_k , $k = 1, \dots, K$, and producing a block of C_s target parameters per pass; the full weight vector is assembled by concatenation. Chunking makes the hypernetwork parameter-efficient but introduces the chunk-induced gradient accumulation pathology addressed in Section 4.5.1.

4.4.2 Functional Regularizer

The functional regularizer of Oswald et al. (2020) is active for all hypernetwork configurations, with regularization strength β treated as a tunable hyperparameter (see Appendix 8).

4.4.3 Target Network and Dropout

Standalone configurations reproduce the target network architecture of Garg et al. (2026) without modification, including two Dropout($p=0.5$) layers in the SimpleCIFARCNN classifier head. Dropout is removed from the target network in all hypernetwork configurations: PyTorch’s `functional_call` propagates the parent module’s `.training` mode to the target submodule, causing stochastic masking across all chunk-generation passes during training. The generative bottleneck ($\phi : d_t + d_c \rightarrow d_h \rightarrow C_s$) and the functional regularizer $\mathcal{L}_{\text{reg}}$ provide sufficient implicit regularization in its place.

4.5 Novel Algorithmic Contributions

4.5.1 Projection-Scoped Normalization

When a hypernetwork spawns K sequential weight-chunks before a single projection step executes, the accumulated gradient vector and the diagonal Fisher entries grow in magnitude proportionally to K , inflating the condition number of the projection kernel $G^\top \hat{F} G$ and causing the matrix inversion in Equation (13) to fail catastrophically for any practically sized target network.

We resolve this with two independent rescaling operations that address distinct sources of ill-conditioning:

1. **Gradient basis orthonormalization.** Each batch of K gradient vectors collected after task t is replaced by the orthonormal factor Q from its thin QR decomposition before insertion into G :

$$[g_{t,1} \mid \cdots \mid g_{t,K}] \xrightarrow{\text{QR}} Q, \quad Q^\top Q = I. \quad (17)$$

QR is chosen over SVD at insertion time because the operation requires only orthonormalization of the incoming batch — the number of columns is preserved, not reduced, so the ordered factorization that SVD provides is unnecessary overhead. SVD is reserved for the overflow handler: when the column count of G exceeds the budget M , the full basis is recompressed as $G \leftarrow U_{:,1:M}$, retaining the M left singular vectors of greatest variance and discarding the rest. This operation applies *at storage time* and affects only the basis G ; the gradient g entering the projection update itself is the unmodified parameter gradient from the model.

2. **Metric-tensor scaling.** Within each projection computation, all Fisher quantities are jointly divided by a single scale factor derived from the maximum entry of the method’s ambient metric F^* :

$$s = \max(\max_i(F^*)_i, 1), \quad \tilde{F} = F / s \text{ for } F \in \{\hat{F}_{\text{old}}, \hat{F}_{\text{new}}, F^*\}, \quad \tilde{\lambda} = \lambda / s, \quad (18)$$

where $F^* = \hat{F}_{\text{new}}$ for FNG, ONG, and FOPNG, and $F^* = F_c$ for iFOPNG. The clamp to 1 prevents division below unity when the Fisher is near-zero. Critically, the damping coefficient λ is rescaled by the same factor, so its ratio to the Fisher magnitude is preserved across tasks regardless of how the Fisher scale drifts as more tasks accumulate. Because the projection kernel A , its pseudo-inverse, and the trust-region denominator are all computed under the same rescaled metric, the direction of v^* is invariant to s ; only the condition number of A is reduced, which is the sole purpose of the operation.

Together, these operations ensure $G^\top \tilde{F} G$ remains well-conditioned regardless of the number of chunks spawned per task. Gradient orthonormalization eliminates column-scale inflation in G ; metric-tensor normalization bounds the Fisher entries to $[0, 1]$, preventing the kernel from becoming numerically singular. Subspace membership is invariant to positive scalar rescaling of basis vectors, so neither operation alters the geometry of the projection.

4.6 Experimental Design

The four sub-research questions demand qualitatively different experimental designs. Sub-RQ1 and Sub-RQ3 require head-to-head benchmark comparisons — projection methods against an unconstrained baseline, and iFOPNG against FOPNG respectively. Sub-RQ2 requires a controlled ablation over normalization conditions. Sub-RQ4 requires a long-horizon comparison of two Fisher accumulation operators on the same inertial combined-metric preconditioner.

4.6.1 Hyperparameter Provenance

STANDALONE TARGET-NETWORK EXPERIMENTS. All hyperparameter values for experiments without a hypernetwork — learning rates, regularization coefficients, gradient budgets, Fisher sample counts, batch sizes, and epoch counts — are taken without modification from Garg et al. (2026) Table 1. iFOPNG uses the same values as FOPNG, since the inertial combined Fisher introduces no hyperparameters beyond those FOPNG already defines. ONG (Yadav et al., 2025) is treated as a composite method: it applies

OGD’s Euclidean projection operator to the natural gradient direction and introduces no hyperparameters beyond those already specified for OGD and FNG; its values therefore follow OGD’s Table 1 entries.

DEVIATION FOR FIRST-TASK INITIALIZATION. We deviated from the original protocol in one specific instance: first-task initialization for MNIST-based benchmarks. While Garg et al. (2026) state they used the method’s own learning rate for first-task SGD training (e.g., $\eta = 10^{-5}$ for FOPNG), our empirical replication of this protocol yielded an initial accuracy of only $\approx 21\%$. As demonstrated by the learning rate convergence sweep detailed in Appendix D (Figure 15), this published rate falls well below the empirical convergence threshold, effectively failing to establish a functional baseline. To ensure a clean initial projection subspace, we standardized the first-task SGD learning rate to 10^{-3} for MNIST-based benchmarks and 10^{-2} for CIFAR-based benchmarks. Beyond this necessary empirical correction, no tuning was performed on the published values.

HYPERNETWORK EXPERIMENTS. Settings not covered by Garg et al. (2026) follow the canonical hypernetwork framework of Oswald et al. (2020); exact values are detailed in Appendix 8. Because computing a full Fisher estimate requires a complete K -chunk generation pass per training sample, estimation is capped at 1,024 samples to maintain computational tractability — a standard approximation consistent with prior CIFAR-10 evaluations (Kirkpatrick et al., 2017). The gradient budget remains $K = 80$ per task, with memory compression handled via QR-and-SVD (Section 4.3). First-task training uses AdamW to seed the memory matrix G and Fisher estimate $\hat{F}$ from a clean geometric basis.

UNIFORM LEARNING RATE CONTROL. To isolate the projection constraint as the sole independent variable in the hypernetwork experiments, all methods use a uniform learning rate of 10^{-3} rather than method-specific rates. This prevents convergence-speed artifacts from conflating the structural effects of the projection geometry.

EARLY STOPPING. Per-task training duration is governed by a single early-stopping rule shared verbatim by every method, baseline and projection alike. Training on a task terminates at the first of three conditions: the average training loss falls below 0.1; ten consecutive epochs elapse without improving the best training loss observed on that task; or the benchmark’s epoch cap (Appendix 8) is reached. A coupled scheduler reduces the learning rate after every three consecutive non-improving epochs, stepping through half-magnitude decrements ($10^{-2} \rightarrow 5 \times 10^{-3} \rightarrow 10^{-3} \rightarrow \dots$).

Upon termination, the parameters of the best-loss epoch — not the final epoch — are restored and carried into the next task. The criterion consults only the training loss, so the stopping rule never observes held-out data; per-task duration is therefore adaptive while introducing no method-specific degree of freedom.

RANDOM SEEDS. All experiments draw from a fixed seed pool. Configurations evaluated on five seeds use $\{42, 111, 811, 1234, 2137\}$; compute-limited configurations use the three-seed subset $\{42, 811, 1234\}$. The seed count for every reported comparison is stated alongside the corresponding result. One deviation applies: on the Split-MNIST hypernetwork benchmark, seed 2137 produced degenerate runs for every projection-based method while leaving the unconstrained baselines unaffected; it was therefore excluded and replaced by seed 314, yielding the five-seed set $\{42, 111, 314, 811, 1234\}$ used uniformly across all methods on this benchmark.

FISHER ACCUMULATION DEFAULT. The EMA decay parameter α appears exclusively in the Sub-RQ4 operator comparison ($\alpha = 0.5$); every other experiment accumulates $\hat{F}_{\text{old}}$ with the element-wise MAX operator, which introduces no additional hyperparameter.

4.6.2 Sub-RQ1 — Constructive Synergy or Architectural Conflict

EXPERIMENTAL DESIGN. Sub-RQ1 investigates whether layering a hard geometric projection on top of a hypernetwork’s soft functional regularizer yields a net benefit, or if the loss of adaptive momentum makes a standard Adam-regularized hypernetwork superior. To separate architectural conflicts from benchmark-specific artifacts, we evaluate across three datasets at varying levels of compression, comparing an unconstrained standalone network (Panel a) against a hypernetwork (Panel b).

BENCHMARK SELECTION. We utilize a three-benchmark progression. **Benchmark 1 (Split-MNIST)** utilizes a deliberately “suffocated” hypernetwork (generator hidden dimension $d_h = 8$) to expose differential effects without hitting the near-perfect accuracy ceilings observed at higher capacities (see Appendix 8, Figure 11). To ensure convergence under these constraints, each task is trained for 15 epochs with a batch size of 64. **Benchmark 2 (Split-CIFAR10)** serves as the primary test, offering genuine inter-task interference and utilizing a standard-capacity hypernetwork trained for 50 epochs. **Benchmark 3 (Split-CIFAR100)** extends this sequence to 10 tasks to test scalability. Comparing performance within and across these panels reveals whether projection methods universally domi-

nate, conditionally succeed based on compression, or systematically fail due to momentum stripping.

4.6.3 Sub-RQ2 — Projection-Scoped Normalization

WHAT WE WANT TO DIFFERENTIATE. Sub-RQ2 asks whether the chunk-induced gradient accumulation pathology is real, characterisable, and resolvable. A single accuracy number cannot answer this: we need direct numerical evidence that the pathology occurs without normalization, that normalization resolves it, and that the resolution does not compromise projection quality.

EXPERIMENTAL DESIGN. iFOPNG is evaluated under three normalization conditions on the Split-CIFAR10 hypernetwork (Benchmark 2), with all other hyperparameters held constant. Split-CIFAR10 is chosen over the CIFAR-100 and MNIST hypernetwork configurations specifically because its chunk size of 6 000 yields $C = 198$ chunks per target network — large enough to produce a measurable pathology, yet small enough that the condition number of A can be computed and logged at every projection step without becoming numerically intractable. Larger chunk counts (e.g. the CIFAR-100 configuration) would make the condition number measurement itself unreliable; smaller counts (e.g. the MNIST suffocated regime) would not stress the pathology sufficiently.

CONDITION 1 — NO NORMALIZATION (NEGATIVE CONTROL). Incoming gradient batches are appended to G without orthonormalization; the Fisher diagonal is used at its raw scale. With 198 chunks accumulated per forward pass, the gradient columns of G and the entries of $\hat{F}$ are expected to reach extreme magnitudes, inflating the condition number of the projection kernel and driving the matrix inversion to NaN. This condition is the negative control that demonstrates the pathology concretely.

CONDITION 2 — GRADIENT ORTHONORMALIZATION ONLY. Each incoming gradient batch is orthonormalized via QR decomposition before insertion into G ; the Fisher diagonal is used at its raw scale. This isolates whether gradient-side conditioning alone is sufficient, or whether Fisher-side scaling is also required.

CONDITION 3 — FULL NORMALIZATION (PROPOSED). Gradient batches are QR-orthonormalized at insertion, and the Fisher terms are rescaled by the maximum entry of the ambient metric F^* within each projection computation. This is the proposed scheme; all other hypernetwork experiments use it.

In addition to accuracy and BWT, the condition number $\kappa(A)$ of the projection kernel is logged before each inversion. A condition number that diverges in Condition 1 but remains bounded in Condition 3 constitutes direct numerical evidence for the pathology and its resolution. All conditions use the Benchmark 2 hypernetwork settings, so any performance difference is attributable to normalization alone.

4.6.4 Sub-RQ3 — Does the Combined Fisher Metric Yield Consistent Retention Improvements

BENCHMARK SELECTION AND RATIONALE. Sub-RQ3 compares iFOPNG against FOPNG across the four standalone benchmarks evaluated by Garg et al. (2026): Permuted-MNIST (5 tasks), Split-MNIST single-head, Split-CIFAR10 multi-head, and Split-CIFAR100 multi-head. All hyperparameters are taken directly from Garg et al.’s Table 1, with the sole exception of the first-task learning rate for MNIST-based benchmarks, where the published value produces a degenerate initialisation; the necessary correction is established empirically in Appendix D (Figure 15). This replication-faithful design ensures that any observed difference between iFOPNG and FOPNG is attributable to the combined metric $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$ rather than to differences in training conditions.

Split-MNIST multi-head and the two hypernetwork configurations (Split-MNIST HN and Split-CIFAR10 HN) were already conducted for Sub-RQ1 and are included here for completeness; they are not part of the Garg et al. replication and carry no additional inferential weight for Sub-RQ3.

4.6.5 Sub-RQ4 — Fisher Accumulation Operator

WHAT WE WANT TO DIFFERENTIATE. The non-decaying maximum $F_{\text{old}} \leftarrow \max(F_{\text{old}}, F_{\text{new}})$ guarantees that any parameter ever deemed important to a past task retains that importance permanently. This is protective for early tasks but monotonic by construction: the inertial metric boundary can only tighten, never relax, so in a sufficiently long sequence the accumulated metric may immobilise the parameters needed to learn later tasks.

EXPERIMENTAL DESIGN. iFOPNG is run under two accumulation operators, all other settings held constant:

MAXIMUM (PROPOSED). $F_{\text{old}} \leftarrow \max(F_{\text{old}}, F_{\text{new}})$ — non-decaying.

EXPONENTIAL MOVING AVERAGE. $F_{\text{old}} \leftarrow (1 - \alpha) F_{\text{old}} + \alpha F_{\text{new}}$ — decaying, allowing historical importance to fade.

The comparison is run on the longest available task sequences — Permuted-MNIST — since the plasticity cost of monotonic accumulation, if present, only becomes observable as the sequence length grows. Per-task accuracy is tracked across the full sequence rather than reported only as a final aggregate: the diagnostic signature of plasticity paralysis is a progressive decline in *newly learned* task accuracy ($a_{t,t}$ falling as t increases), distinct from forgetting, which appears as decline in *previously learned* task accuracy. If the maximum operator shows superior early-task retention but degrading $a_{t,t}$ in late tasks, while the moving average shows the opposite trade, the two failure modes are cleanly separated and Sub-RQ4 is answered in full.

5 RESULTS

For each sub-research question, we describe the experimental scope, present the key quantitative findings, and highlight the most important patterns; detailed interpretation is reserved for the Discussion. *Average accuracy* (A_T) is the primary evaluation metric, and *backward transfer* (BWT) is reported as a secondary measure of forgetting. Standalone target-network results that establish the reference behaviour of projection methods in their original, non-hypernetwork setting are provided in Appendix B (Figures 7–9).

5.1 Sub-RQ1: Constructive Synergy or Architectural Conflict?

Three hypernetwork regimes are evaluated: a standard-capacity Split-CIFAR10 network, an extremely compressed Split-MNIST network (suffocated regime, $d_h = 8$), and a ten-task Split-CIFAR100 network.

STANDARD HYPERNETWORK — SPLIT-CIFAR10. Figure 1 presents average accuracy and BWT for all eight methods on Split-CIFAR10 with a standard-capacity hypernetwork and full normalization enabled (see Section 5.2). Plain SGD at 84.1% establishes the unconstrained reference, and Adam at 90.4% defines the upper frontier. iFOPNG leads all projection methods at 85.6%, the only one to surpass SGD. The remaining methods fall below the SGD reference: EWC at 83.1%, FNG at 81.1%, OGD at 79.6%, FOPNG at 77.9%, and ONG at 71.4%. The result confirms a partial constructive synergy: hard projection combined with the functional regularizer yields iFOPNG’s marginal improvement over SGD, but projection alone does not lift most methods above the unconstrained baseline, and none match Adam, which leads by 4.5 pp over the best projection method.

The BWT values reveal an instructive asymmetry: Adam has effectively zero backward transfer ($\text{BWT} \approx 0.000$), while all projection methods exhibit small but nonzero forgetting (iFOPNG: -0.013 ; FOPNG: -0.015).

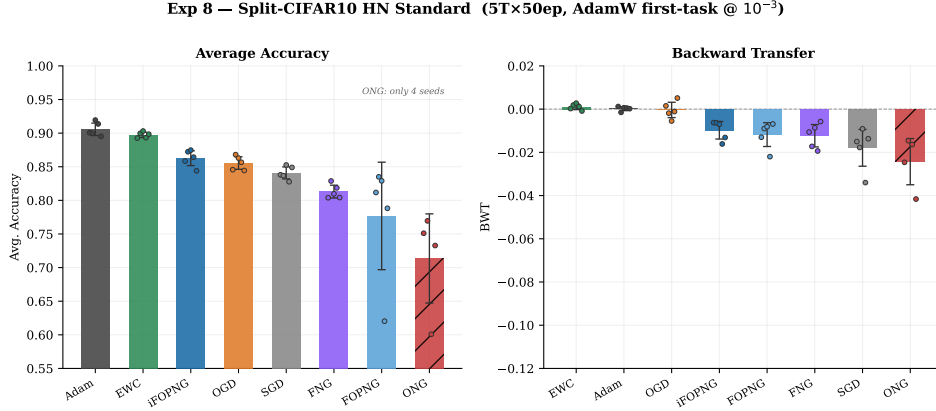


Figure 1: **Split-CIFAR10 hypernetwork (5 tasks, 50 epochs per task, AdamW first-task initialisation at 10^{-3})**. Left: average accuracy (bars show mean over seeds; circles show individual seeds). Right: backward transfer. Methods are sorted by average accuracy on the left panel. Adam defines the projection-free upper frontier; SGD defines the unconstrained reference at 84.1%. iFOPNG is the only projection method to exceed SGD; the remaining projection variants fall below it. All experiments use full normalization (Section 5.2).

SUFFOCATED REGIME — SPLIT-MNIST, $d_h = 8$. The embedding dimension $d_h = 8$ was selected via a three-point sweep over $d_h \in \{4, 8, 16\}$, presented as an ablation in Figure 11. That sweep shows the projection-Adam gap is largest at $d_h = 8$ (23.5 pp), falls to 14.0 pp at $d_h = 4$ where seed variance becomes extreme and results near-random, and shrinks to 9.4 pp at $d_h = 16$ as the projection methods partially recover. $d_h = 8$ was retained as the most diagnostically informative level: the failure mode is severe and replicable, but the network has sufficient capacity for the projections to converge consistently.

Figure 2 presents the full method comparison at $d_h = 8$. Adam leads at 93.4%. The Fisher-free projection method OGD is the strongest projection baseline at 86.6%, followed by EWC at 84.5% and SGD at 79.8%. The Fisher-based methods fall sharply below: iFOPNG at 69.9%, FOPNG at 59.9%, and FNG at 54.8% — gaps of 23–39 pp against Adam.

The BWT panel reveals the mechanism underlying this collapse. iFOPNG, FOPNG, and FNG all achieve near-zero backward transfer ($\text{BWT} \approx 0.00$), while Adam has small negative BWT (-0.015) and SGD forgets more substantially (-0.13).

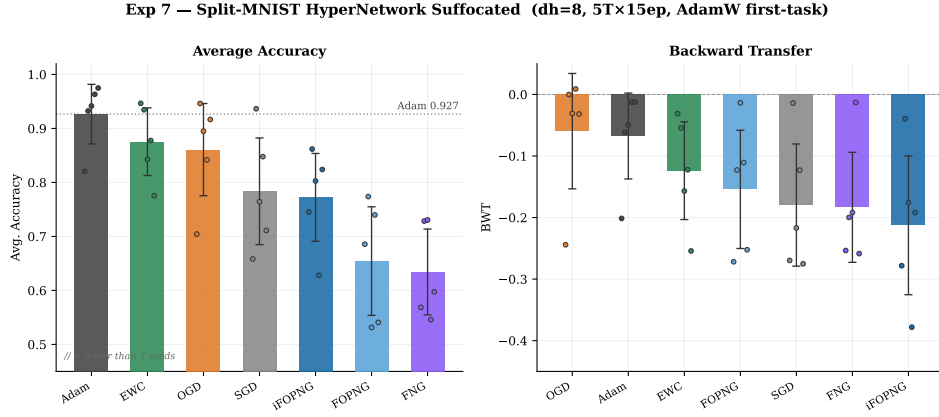


Figure 2: **Split-MNIST SH hypernetwork, suffocated regime ($d_h = 8$, 5 tasks, 15 epochs per task, AdamW first-task initialisation)**. Left: average accuracy (bars show mean over seeds; circles show individual seeds; note that several methods completed fewer than three seeds). Right: backward transfer. The dotted line marks Adam’s mean accuracy (93.4%). Fisher-based projection methods (iFOPNG, FOPNG, FNG) achieve near-zero BWT but much lower accuracy than Adam, revealing a plasticity failure rather than a retention failure in the extreme compression regime. The Fisher-free OGD is the best-performing projection method at 86.6%.

EXTENDED TASK COUNT — SPLIT-CIFAR100. At ten tasks on Split-CIFAR100 (Appendix Figure 14), EWC unexpectedly leads at 45.5%, with Adam second at 42.0% and iFOPNG third at 37.8% (random baseline: 10.0%). This partial rank reversal, in which soft regularization overtakes both Adam and gradient projection, differs from the five-task CIFAR10 pattern and is addressed in the Discussion.

5.2 Sub-RQ2: Normalization Resolves the Gradient Pathology

This subsection reports the three-condition normalization ablation (Split-CIFAR10 hypernetwork; 197 chunks, ≈ 6000 output neurons; unnormalized conditions ran for 10 epochs per task with early stopping, full normalization for 50 epochs). Figure 3 traces the $\log_{10}$ condition number of the projection kernel $G^\top \hat{F} G$ over training steps; Figure 4 shows the resulting average accuracy and BWT under each condition.

WITHOUT NORMALIZATION. The right panel of Figure 3 shows the condition number reaching $10^{10.8}$ at the T2→T3 task transition for the most-affected seed, stabilizing near $10^{8.8}$ – $10^{9.0}$ for two others. At this scale, numerical inversion of the projection kernel is unreliable, effectively zeroing the projected gradient so that the method operates as an unconstrained optimizer with unstable Fisher scaling. The result is 72.2% average accuracy

and $\text{BWT} = -0.070$ — worse than plain SGD (84.1%) despite identical hyperparameters. The pathology therefore not only eliminates the benefit of projection but actively degrades performance below the projection-free baseline.

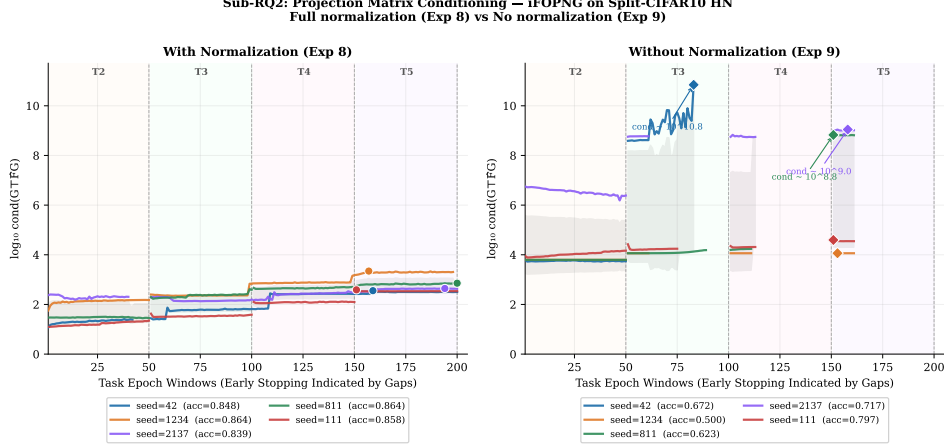


Figure 3: Projection kernel condition number over training — iFOPNG on Split-CIFAR10 HN. Each line is one seed (legend shows final average accuracy). Task boundaries are marked by vertical dashed lines; gaps indicate early stopping. Left (full normalization): $\log_{10} \kappa(G^\top \hat{F} G)$ remains bounded within $[10^{1.3}, 10^{3.3}]$ throughout. Right (no normalization): condition number spikes to $10^{10.8}$ at the T2→T3 transition, causing projection failure. Diamond markers indicate end-of-task condition number values.

GRADIENT-ONLY NORMALIZATION. Applying unit- ℓ_2 column normalization to G alone, without rescaling $\hat{F}$, fails to resolve the pathology. Average accuracy falls further to 68.9% ($\text{BWT} = -0.060$) and variance across seeds remains high (Figure 4). As the left panel of Figure 3 illustrates, the raw Fisher magnitudes from the chunked backward passes still span several orders of magnitude, preserving the ill-conditioning of $G^\top \hat{F} G$ even when G itself is normalized. Partial normalization is therefore insufficient.

FULL NORMALIZATION. Combining column-wise ℓ_2 normalization of G with absolute-maximum scaling of $\hat{F}$ fully resolves the pathology. The left panel of Figure 3 shows $\log_{10} \kappa(G^\top \hat{F} G)$ bounded within $[10^{1.3}, 10^{3.3}]$ across all seeds and task transitions — a reduction of more than seven orders of magnitude. Figure 4 shows the result: average accuracy recovers to 85.8% and BWT improves to -0.013 , a 5.7 pp reduction in forgetting relative to no normalization. The improvement is consistent across all five

seeds. Full normalization is adopted for all hypernetwork experiments in this thesis.

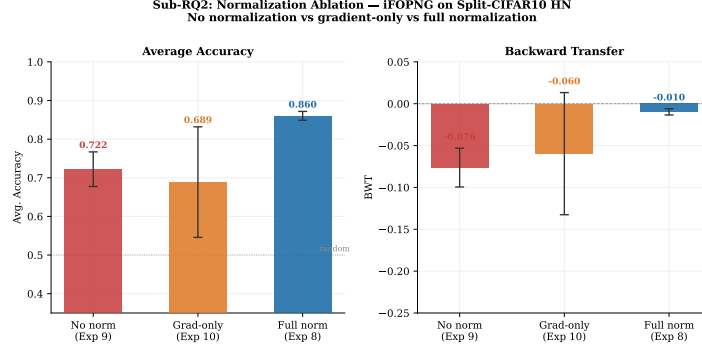


Figure 4: **Normalization ablation — iFOPNG on Split-CIFAR₁₀ HN (Sub-RQ₂).** Three conditions: no normalization, gradient-only normalization, and full normalization. Left: average accuracy with mean annotated on each bar. Right: backward transfer. Circles show individual seeds; bars show the mean. The dotted line on the left panel marks random-chance performance (50% for binary per-task classification).

5.3 Sub-RQ₃: Parameter Inertia via the Combined Fisher Metric

Figure 5 presents mean average accuracy for iFOPNG and FOPNG across all seven benchmarks.

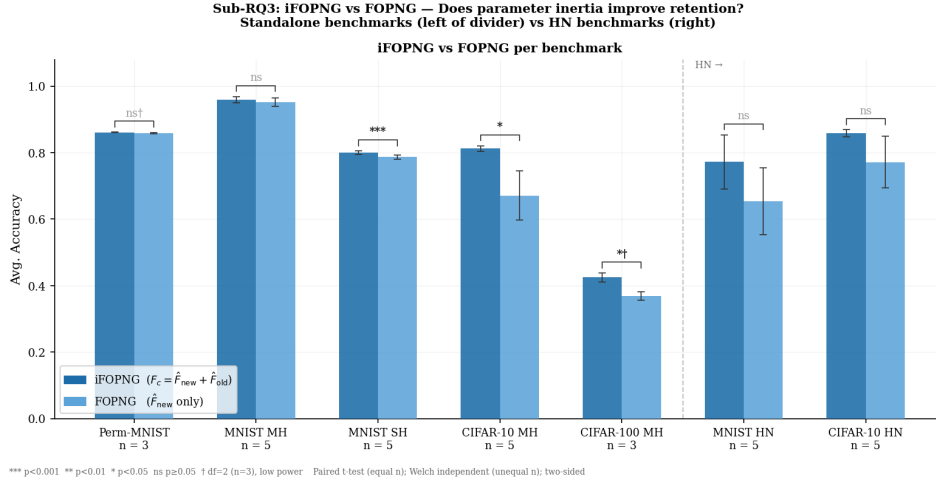


Figure 5: **iFOPNG vs. FOPNG across seven benchmarks (Sub-RQ₃).** Grouped bars show mean average accuracy; circles show individual seeds; error bars show ± 1 standard deviation. The dashed vertical line separates standalone (left) from hypernetwork (right) settings. iFOPNG uses the combined metric $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$; FOPNG uses only $\hat{F}_{\text{new}}$.

On the two simplest benchmarks — Permuted-MNIST and Split-MNIST multi-head — the two methods are nearly indistinguishable (86.1% vs. 85.8% at $n = 3$ seeds, and 95.9% vs. 95.2% at $n = 5$ seeds each; paired comparisons on the four common seeds are non-significant, $t(3) = 1.72$, $p = 0.185$), consistent with low inter-task Fisher overlap on near-ceiling tasks.

Split-MNIST single-head provides a cleaner test of the inertial combined-metric preconditioner under output-level interference. Using the task-incremental shared-head protocol (see Related Work, Section 3.2), iFOPNG reaches 80.0% versus FOPNG’s 78.7%, a gap of +1.3 pp that is statistically significant ($t(4) = 24.07$, $p < 0.001$, $d_z = 10.76$). Throughout, $d_z = \bar{D}/s_D$ denotes the paired effect size — the mean per-seed difference divided by its sample standard deviation — so large values indicate seed-to-seed consistency of the improvement rather than its raw magnitude. The full per-task trajectory and BWT evolution are provided in Appendix B (Figure 10).

A clear separation emerges on harder benchmarks. On Split-CIFAR10 multi-head, iFOPNG reaches 81.2% versus FOPNG’s 67.4%, a gap of 13.8 pp that is significant under a paired t -test ($t(4) = 3.54$, $p = .024$, $d_z = 1.58$, $n = 5$ seeds). On Split-CIFAR100 multi-head the advantage is 5.6 pp (42.7% vs. 37.1%; $t(2) = 6.08$, $p = .026$, $n = 3$ seeds). Detailed per-task trajectories for these standalone references are provided in Appendix B (Figures 8 and 9).

The advantage carries over into the hypernetwork settings. On Split-MNIST HN, seed 2137 produced a degenerate run for both methods (10.7% each, indicating a failed run on this configuration rather than a property of either method); it was excluded and replaced by seed 314, preserving $n = 5$. On this set, iFOPNG reaches 77.2% versus FOPNG’s 65.4%, an 11.8 pp gap ($t(4) = 1.61$, $p = .183$). On Split-CIFAR10 HN the gap is 8.6 pp (85.8% vs. 77.2%; $t(4) = 2.18$, $p = .095$, $n = 5$ seeds). Neither hypernetwork comparison reaches statistical significance, reflecting the elevated inter-seed variance of this setting. Across all seven benchmarks, the retention advantage of iFOPNG scales with task difficulty and output-level interference.

5.4 Sub-RQ4: Fisher Accumulation Operator

This subsection compares two strategies for accumulating F_{old} across tasks under the iFOPNG combined Fisher metric: the element-wise maximum operator (MAX) and an exponential moving average (EMA). Both are evaluated on a twenty-task Permuted-MNIST sequence with five random seeds. Figure 6 presents the average accuracy and backward transfer

trajectories over all twenty tasks; the per-seed statistical comparison is provided in Appendix Figure 17.

Config 14 & 15: EMA vs MAX Fisher Accumulation (Averaged Across Seeds)

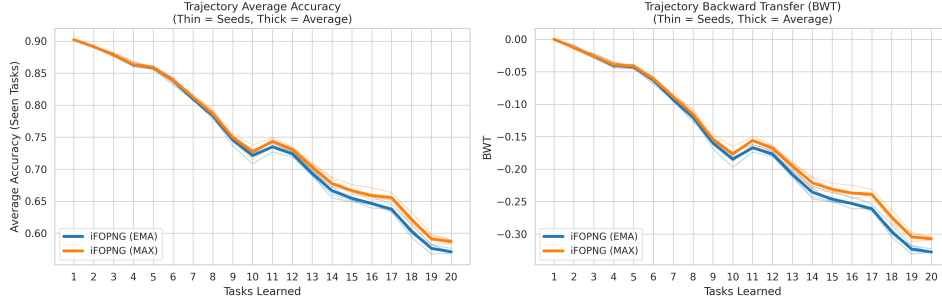


Figure 6: **EMA vs. MAX Fisher accumulation trajectories on 20-task Permuted-MNIST (Sub-RQ4), $n = 5$ seeds.** Thick lines show the seed-averaged trajectory; thin lines show individual seeds. Left: average accuracy over tasks seen so far, evaluated after each task is trained. Right: backward transfer over the same sequence. MAX leads EMA by a consistent margin across all five seeds; both strategies produce closely tracking trajectories throughout the full twenty-task sequence.

A paired t -test on final average accuracy establishes MAX as the superior operator: $t(4) = 14.43$, $p < 0.001$, $d_z = 6.45$, with MAX leading by 1.50–1.65 pp across every seed pair without exception (Appendix Figure 17). The result is statistically unambiguous and replicates across all five seeds. In absolute terms the margin is practically negligible — 1.60 pp on a benchmark where both variants decline by more than 30 pp over the full sequence — but the direction is consistent and the effect size large, leaving no ambiguity about which operator to prefer.

The MAX operator is therefore preferred as the simpler, marginally better, and hyperparameter-free alternative.

6 DISCUSSION

The overarching goal was to determine whether gradient projection methods can be fruitfully integrated into chunked hypernetworks for continual learning, and to identify the structural conditions under which this integration succeeds or fails.

6.1 Sub-RQ1: Constructive Synergy or Architectural Conflict?

Projection methods do not outperform a well-tuned regularized hypernetwork optimized via Adam. The gap of 4.5 pp at task 5 in favor of Adam is

consistent across seeds and widens substantially in the suffocated regime ($d_h = 8$, Split-MNIST SH), where it reaches 20–40 pp. Critically, the BWT data reveal that in the suffocated regime this is a *plasticity* failure, not a *retention* failure: Fisher-based projection methods achieve near-zero backward transfer alongside collapsed accuracy, meaning the curvature-weighted constraint exhausts the available gradient budget for new task learning rather than permitting forgetting of prior ones. OGD’s relative resilience at $d_h = 8$ is consistent with this reading: without Fisher weighting its subspace constraint is less aggressive, leaving more gradient capacity for forward transfer. We attribute the broader gap to a fundamental architectural incompatibility. Projection methods operate by modifying the raw gradient before the optimizer state update, which means the first-order momentum estimate and the adaptive scaling factor maintained by Adam are effectively discarded at each projected step. In the highly nonlinear, compressed weight space of a hypernetwork meta-generator, momentum is not merely a convergence accelerant — it is a structural regularizer that prevents the optimizer from committing prematurely to poorly-conditioned gradient directions. Re-initializing the optimizer’s internal state at each step destabilizes the generative manifold and reduces the quality of the generated weights, particularly under extreme bottleneck compression. This finding is consistent with [Hu, Yu, Cheng, Liu, and Song \(2026\)](#), who demonstrate that gradient modification methods interact adversely with Adam’s internal state under certain curvature regimes, and aligns with the broader observation that projection-based continual learning methods were originally designed for SGD-like updates.

The ten-task Split-CIFAR100 result introduces a partial rank reversal that complicates the Sub-RQ1 conclusion: EWC unexpectedly leads at 45.5%, overtaking both Adam (42.0%) and iFOPNG (37.8%). As the number of tasks grows, the gradient subspace spanned by G becomes an increasingly coarse approximation to the full historical interference manifold, while the functional regularizer grows more informative with each additional stored weight snapshot. Whether this effect persists at the twenty-task scale, and whether an expanded memory budget would recover the projection advantage, remains to be investigated.

FIRST-TASK INITIALIZATION AND GRADIENT CONTAMINATION. Standard Adam couples weight decay into the gradient as an additive ℓ_2 term, so both G and $\hat{F}$ collected at task 1 inherit a component aligned with weight magnitude rather than task loss curvature, contaminating the projection subspace. AdamW’s decoupled parameter-shrinkage step avoids this: it consistently produced lower first-task trajectory variance and a cleaner initial projection basis. For any system bootstrapping a projection memory from

first-task gradients, the decoupled optimizer should be preferred at that stage.

6.2 Sub-RQ2: Projection-Scoped Normalization Resolves the Pathology

The fundamental problem is purely numerical rather than conceptual: the intersection of chunked hypernetworks and Fisher-weighted gradient projection is geometrically coherent but arithmetically fragile without explicit local rescaling. This normalization is therefore a necessary engineering precondition for any method that uses the Fisher metric within the projection algebra — specifically FNG, FOPNG, and iFOPNG. OGD, which projects orthogonally without Fisher weighting, maintains $G^\top G = I$ exactly through QR orthonormalization and is structurally immune to the inversion collapse; its competitive performance on the suffocated regime (Section 5.1) is consistent with this immunity. Severity scales with chunk count C and training duration: gradient magnitudes accumulate as $C \cdot g_{\text{unit}}$, Fisher as $C^2 \cdot F_{\text{unit}}$, and the unnormalized projection matrix inflates by order C^4 per task. At sufficient scale the collapse is categorical rather than gradational — runs crash or converge to random-chance accuracy rather than merely underperforming, as the elevated failure rate on the ten-task Split-CIFAR100 hypernetwork illustrates even after normalization is applied — marking normalization as a structural stability prerequisite, not a performance tuning.

6.3 Sub-RQ3: Parameter Inertia via the Combined Fisher Metric

iFOPNG outperforms FOPNG in all seven settings, confirming that the combined metric $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$ yields consistent retention improvements. The advantage is positive in every case and seed-consistent: even the smallest gap, on Split-MNIST single-head (+1.3 pp), is statistically unambiguous ($p < 0.001$, $d_z = 10.76$), indicating that the inertial term reliably suppresses over-aggressive natural gradient steps at the parameters that matter most to prior tasks whenever backbone and output-level interference co-occur simultaneously. The magnitude varies across benchmarks but the direction does not: Sub-RQ3 is answered affirmatively. iFOPNG does not close the gap to Adam, confirming that the inertial combined metric improves retention by weighting historically important coordinates more heavily, but cannot recover the adaptive momentum that Adam leverages for efficient navigation of the generative manifold.

6.4 Sub-RQ4: Fisher Accumulation Operator

On a twenty-task Permuted-MNIST sequence, the element-wise maximum (MAX) accumulation strategy systematically outperformed the exponential moving average (EMA). While the 1.60 pp advantage is small, it is statistically unambiguous ($p < 0.001$). Crucially, the hypothesized plasticity cost of MAX’s monotonic growth did not materialize. Because our normalization scheme rescales both Fisher terms, the projection becomes insensitive to absolute scale, relying solely on relative parameter distributions. Therefore, MAX remains the definitive practical choice: it is simpler, hyperparameter-free, and marginally superior, though highly heterogeneous task distributions could potentially introduce divergence.

6.5 Limitations

Several limitations should be acknowledged.

First, the hypernetwork results presented here are restricted to Split-MNIST and Split-CIFAR10 with five tasks. Longer task sequences may produce different relative orderings: in particular, the advantage of hard projection constraints over soft functional regularization may grow with sequence length, as soft penalties are known to suffer from compounding output drift (Farquhar & Gal, 2018), while the projection subspace scales with the gradient budget rather than the task count. The preliminary ten-task Split-CIFAR100 result, in which EWC overtakes both Adam and iFOPNG, is suggestive but insufficient to establish a trend; a systematic sweep over task count is warranted.

Second, the results are specific to the constrained-capacity regimes examined here. For Split-MNIST, a standard-capacity hypernetwork solves the benchmark near-trivially regardless of the continual learning method applied; the suffocated setting ($d_h = 8$) is deliberately chosen to place the architecture below saturation, where method differences are measurable. For Split-CIFAR10, the convolutional target network is large enough that the hypernetwork already operates under a natural capacity constraint, and meaningful method separation emerges without artificial bottlenecking. Whether the projection-hypernetwork interaction studied here generalises to harder benchmarks requiring larger target architectures — where the hypernetwork must generate weights for deeper convolutional or attention-based networks and the task itself is sufficiently complex to prevent ceiling effects — remains an open question. In such settings, the gradient accumulation pathology documented here would be substantially more severe: chunk counts scale with the target parameter count, so the condition num-

ber blowup $\sim C^4$ identified here represents a lower bound on the numerical challenge at scale.

Third, the proposed normalization scheme—QR orthonormalization combined with absolute-maximum Fisher scaling—is fundamentally heuristic in nature. While it successfully eliminates the numerical pathology of condition number collapse, it applies isotropic rescaling to the projection basis. This likely discards important geometric properties of the Fisher metric, temporarily divorcing the subspace constraint from the true curvature of the parameter space. A mathematically principled formulation that stabilizes the projection algebra while strictly preserving the theoretical guarantees of Fisher-based methods remains an open question for future work.

7 CONCLUSION

This thesis investigated integrating gradient projection methods into chunked hypernetworks for continual learning. We primarily diagnosed the architectural boundaries and numerical pathologies governing this core integration. Alongside these fundamental structural findings, we also evaluated adjacent methodological optimizations regarding parameter inertia and Fisher matrix accumulation.

SUB-RQ1 — PARTIAL SYNERGY AND THE MOMENTUM BOTTLENECK. The synergy between hard gradient projection and soft functional regularization is constructive but bounded. While iFOPNG outperforms unconstrained SGD in standard-capacity settings, standalone regularized hypernetworks optimized via Adam remain vastly superior.

This performance gap stems from an architectural conflict, not projection geometry. Under extreme compression ($d_h = 8$), Fisher-based projection methods suffer a severe *plasticity* failure, exhausting the gradient budget and halting new task acquisition. Because projection modifies raw gradients before the optimizer updates, it discards adaptive momentum. In hypernetworks, this momentum acts as a crucial structural regularizer; without it, the network commits prematurely to poorly-conditioned gradient directions.

Ultimately, the loss of adaptive momentum is the binding constraint. Integrating gradient projection into hypernetworks will require projection-compatible adaptive optimizers, rather than post-hoc gradient modifications (Hu et al., 2026).

SUB-RQ2 — NORMALIZATION IS A NECESSARY CONDITION. Without explicit local rescaling, projection methods are numerically incompatible

with chunked hypernetworks: matrix inversion in the projection kernel collapses completely, degrading performance below plain SGD. QR orthonormalization of the gradient basis combined with absolute-maximum Fisher scaling resolves the pathology entirely. This normalization scheme is a necessary engineering precondition for any future combination of gradient projection with chunked meta-learners, regardless of which projection algebra is employed.

SUB-RQ3 — PARAMETER INERTIA VIA COMBINED FISHER METRIC. Replacing the task-specific Fisher metric with the combined inertial metric (F_c) systematically improves retention as task difficulty and output-level interference increase. On simple benchmarks, both methods are statistically indistinguishable. On harder standalone benchmarks like Split-CIFAR10, iFOPNG provides a clear, statistically significant advantage (+13.8 pp, $p < 0.05$). This substantial raw advantage carries over into the hypernetwork settings, yielding performance gaps of +11.8 pp and +8.6 pp. However, these specific hypernetwork improvements do not reach statistical significance due to elevated inter-seed variance. Ultimately, integrating parameter inertia yields a consistent retention advantage that scales directly with benchmark complexity.

SUB-RQ4 — MAX ACCUMULATION IS THE CLEAR PRACTICAL CHOICE. Addressing the trade-off between long-horizon retention and plasticity, the element-wise maximum (MAX) operator proved systematically superior to an exponential moving average (EMA). While the absolute performance advantage is small (~ 1.60 pp) on a twenty-task sequence, it is statistically unambiguous across all seeds. Crucially, the hypothesized plasticity cost associated with the monotonic growth of the MAX operator did not materialize over the extended task sequence, likely because the chosen normalization scheme minimizes sensitivity to absolute scale. Therefore, MAX emerges as the definitive practical choice: it avoids plasticity degradation while remaining a simpler, hyperparameter-free, and marginally superior alternative.

IMPLICATIONS FOR THE FIELD. The results establish a clear empirical boundary for the current generation of gradient projection methods: they can be made numerically stable in chunked hypernetwork settings via local rescaling, and their Fisher geometry can be improved via the combined Fisher metric F_c , but the momentum incompatibility prevents them from reaching the performance frontier of adaptive unconstrained optimizers in generative parameter spaces. The normalization contribution is directly applicable to any method that maintains a gradient subspace memory

alongside a chunked weight generator. The iFOPNG contribution extends naturally to any continual learning setting where Fisher overlap accumulates across tasks. Together, they narrow the gap between projection-based and regularization-based approaches, and provide a principled foundation for the next step: a projected adaptive optimizer that respects both the geometric constraints of the projection and the curvature structure that adaptive moment estimates capture.

FUTURE DIRECTIONS. The most immediate extension is developing projection-compatible adaptive optimizers that maintain momentum across constrained updates (Hu et al., 2026). Additionally, the heuristic QR normalization, FIM max scaling scheme must be replaced with a principled reformulation that integrates chunk-Jacobian curvature directly into the projection algebra. Finally, future work should systematically evaluate these interactions on extended task sequences, where soft regularization penalties may degrade, and on larger target architectures, where the documented condition number blowup will require more robust mitigation.

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The algorithmic framework of this thesis builds directly on the Fisher-Orthogonal Projected Natural Gradient method of Garg et al. (2026), whose derivations and design served as the primary technical foundation for iFOPNG. The hypernetwork architecture, weight-chunking scheme, and functional regularizer are due to Oswald et al. (2020) and are adopted here without modification.

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8 APPENDIX

APPENDIX A: IFOPNG THEOREM AND PROOF

A.1 Notation and Standing Conventions

Throughout this appendix, p denotes the number of trainable parameters and m the number of stored gradient directions, with $m \ll p$. The vector $g \in \mathbb{R}^p$ is the current mini-batch gradient, and $G \in \mathbb{R}^{p \times m}$ is the memory matrix whose columns are the stored historical gradient directions (orthonormalised at insertion time via QR decomposition; the proof makes no use of this property — see Remark 8). F_{new} and F_{old} denote the current-task and accumulated diagonal empirical Fisher estimates. Both are positive semi-definite by construction: the diagonal entries of F_{new} are averages of squared score components, and F_{old} is an element-wise maximum of such estimates, an operation that preserves entry-wise non-negativity. The combined metric is $F_c := F_{\text{new}} + F_{\text{old}}$.

Diagonal Fisher tensors are stored as vectors and act by element-wise multiplication; in all displays below they are identified with the corresponding $p \times p$ diagonal matrices, which are in particular symmetric. For a symmetric positive-definite matrix S , we write $\langle w, z \rangle_S := w^\top S z$ for the induced inner product and $\|w\|_S := \sqrt{w^\top S w}$ for the induced norm; $\text{Col}(M)$ denotes the column space of a matrix M . Two derived quantities recur throughout:

$$B := F_{\text{old}} G \in \mathbb{R}^{p \times m}, \quad A := B^\top F_c^{-1} B = G^\top F_{\text{old}} F_c^{-1} F_{\text{old}} G \in \mathbb{R}^{m \times m}. \quad (19)$$

A.2 Assumptions

Assumption 1 (Positive-definite combined metric). $F_c \succ 0$; equivalently, $(F_{\text{new}})_i + (F_{\text{old}})_i > 0$ for every coordinate i .

Since both summands are positive semi-definite, Assumption 1 can fail only on coordinates whose score component is identically zero under both Fisher estimates — parameters that influenced neither the current task nor any previous one. The implementation inverts $F_c + \hat{\lambda}I$ with damping $\hat{\lambda} > 0$, which is positive definite unconditionally; the theorem below treats the exact ($\lambda \rightarrow 0$) problem, and Remark 8 addresses the damped form used in practice.

Assumption 2 (Projection kernel rank). The kernel matrix A is invertible. By Lemma 1 below, this is equivalent to $\text{rank}(F_{\text{old}} G) = m$: no non-trivial linear combination of stored directions is annihilated by the accumulated Fisher.

QR orthonormalisation at insertion guarantees $\text{rank}(G) = m$, but rank can still be lost after weighting by F_{old} if some stored direction is supported entirely on coordinates with $(F_{\text{old}})_i = 0$. This is the only mechanism of rank loss; the damped kernel $A + \hat{\lambda} I_m$ used in the implementation is invertible unconditionally (Remark 8).

Assumption 3 (Non-degenerate residual). $Pg \neq 0$, where $P := I - BA^{-1}B^\top = I - F_{\text{old}} G A^{-1} G^\top F_{\text{old}}$.

Assumption 3 is required only for the trust-region normalisation in Step 5 of the proof. The degenerate event $Pg = 0$ can occur only when g lies in the protected image $\text{Col}(F_{\text{old}} G)$, and the convention $v = 0$ is adopted in that case (Remark 8).

A.3 Auxiliary Results

Lemma 1 (Gram structure of the projection kernel). *Under Assumption 1, the matrix $A = B^\top F_c^{-1} B$ is symmetric positive semi-definite, and the following are equivalent: (i) A is invertible; (ii) $A \succ 0$; (iii) $\text{rank}(B) = m$.*

Proof. $F_c^{-1} \succ 0$ admits a positive-definite square root $F_c^{-1/2}$. Symmetry is immediate from $A^\top = B^\top (F_c^{-1})^\top B = A$. For any $x \in \mathbb{R}^m$,

$$x^\top A x = x^\top B^\top F_c^{-1} B x = \|F_c^{-1/2} B x\|_2^2 \geq 0, \quad (20)$$

so $A \succeq 0$, with equality if and only if $F_c^{-1/2} B x = 0$, i.e. $Bx = 0$ since $F_c^{-1/2}$ is invertible. Hence $A \succ 0$ holds exactly when $Bx = 0$ implies $x = 0$, which is (iii). Finally, a symmetric positive semi-definite matrix is invertible if and only if its smallest eigenvalue is strictly positive, i.e. if and only if it is positive definite, giving (i) $\Leftrightarrow$ (ii). $\square$

Lemma 2 (Strict convexity of the reduced objective). *Under Assumptions 1–2, the function*

$$J(\alpha) := g^\top F_c g - 2g^\top F_{\text{old}} G \alpha + \alpha^\top A \alpha, \quad \alpha \in \mathbb{R}^m, \quad (21)$$

is strictly convex and attains its unique global minimum at $\alpha^ = A^{-1} G^\top F_{\text{old}} g$.*

Proof. J is a quadratic with Hessian $\nabla^2 J = 2A \succ 0$ by Lemma 1, hence strictly convex. The first-order condition $\nabla J(\alpha) = -2G^\top F_{\text{old}} g + 2A\alpha = 0$ has the unique solution $\alpha^* = A^{-1} G^\top F_{\text{old}} g$, and for a strictly convex differentiable function the unique stationary point is the unique global minimiser. $\square$

A.4 Main Result

Theorem 1 (iFOPNG Update). *Let Assumptions 1–3 hold, and consider the two-stage construction*

$$\begin{aligned} (i) \quad u^* &= \arg \min_{u \in \text{Col}(F_{\text{old}}G)} \|g - F_c^{-1}u\|_{F_c}^2, \\ (ii) \quad v &= c F_c^{-1}(g - u^*), \quad \text{with } c < 0 \text{ chosen such that } \|v\|_{F_c} = \eta. \end{aligned} \quad (22)$$

When $F_{\text{old}} \succ 0$, the constraint set in (i) coincides with $\{u : F_{\text{old}}^{-1}u \in \text{Col}(G)\}$, the form quoted in the main text; the formulation via $\text{Col}(F_{\text{old}}G)$ extends it to singular F_{old} . Then u^* exists and is unique, and the resulting update is Equation (16) of the main text:

$$v^* = -\eta \frac{F_c^{-1}Pg}{\sqrt{(Pg)^\top F_c^{-1}Pg}}, \quad P = I - F_{\text{old}} G A^{-1} G^\top F_{\text{old}}. \quad (23)$$

Proof. Step 1 (parameterisation of the constraint set). By definition, $\text{Col}(F_{\text{old}}G) = \{F_{\text{old}}G\alpha : \alpha \in \mathbb{R}^m\} = \{B\alpha : \alpha \in \mathbb{R}^m\}$. Under Assumption 2 and Lemma 1, $\text{rank}(B) = m$, so the map $\alpha \mapsto B\alpha$ is injective: the parameterisation is a bijection between $\mathbb{R}^m$ and the constraint set, and uniqueness of the minimiser in α will transfer to uniqueness in u . When $F_{\text{old}} \succ 0$, the substitution $u = B\alpha \Leftrightarrow F_{\text{old}}^{-1}u = G\alpha \in \text{Col}(G)$ recovers the main-text form of the constraint.

Step 2 (expansion of the objective). For arbitrary $u \in \mathbb{R}^p$, using the symmetry of F_c and $F_c^{-1}F_c = I$,

$$\begin{aligned} \|g - F_c^{-1}u\|_{F_c}^2 &= (g - F_c^{-1}u)^\top F_c (g - F_c^{-1}u) \\ &= g^\top F_c g - 2g^\top F_c F_c^{-1}u + u^\top F_c^{-1}F_c F_c^{-1}u \\ &= g^\top F_c g - 2g^\top u + u^\top F_c^{-1}u. \end{aligned} \quad (24)$$

Substituting $u = B\alpha$ and using $F_{\text{old}}^\top = F_{\text{old}}$, the linear term becomes $-2g^\top B\alpha = -2g^\top F_{\text{old}}G\alpha = -2(G^\top F_{\text{old}}g)^\top \alpha$ and the quadratic term becomes $\alpha^\top B^\top F_c^{-1}B\alpha = \alpha^\top A\alpha$. The reduced objective is therefore exactly the function $J(\alpha)$ of Lemma 2; its first term does not depend on α .

Step 3 (unique minimiser). By Lemma 2, J is strictly convex and its unique global minimiser solves the normal equations

$$A\alpha^* = G^\top F_{\text{old}}g, \quad \text{i.e.} \quad \alpha^* = A^{-1}G^\top F_{\text{old}}g. \quad (25)$$

Step 4 (back-substitution and the projection operator). Substituting α^* into the parameterisation,

$$u^* = B\alpha^* = F_{\text{old}}G A^{-1}G^\top F_{\text{old}}g, \quad \text{hence} \quad g - u^* = (I - F_{\text{old}}G A^{-1}G^\top F_{\text{old}})g = Pg. \quad (26)$$

P is symmetric, since A^{-1} is symmetric by Lemma 1. By Step 1, u^* is the unique solution of stage (i).

Step 5 (well-posedness of the trust-region normalisation). For $v = c F_c^{-1} P g$,

$$\|v\|_{F_c}^2 = c^2 (F_c^{-1} P g)^\top F_c (F_c^{-1} P g) = c^2 (P g)^\top F_c^{-1} F_c F_c^{-1} (P g) = c^2 (P g)^\top F_c^{-1} (P g). \quad (27)$$

The scalar $(P g)^\top F_c^{-1} (P g)$ is a quadratic form in the positive definite matrix F_c^{-1} , hence non-negative, and strictly positive under Assumption 3; the square root below is therefore real and non-zero.

Step 6 (magnitude and sign). The constraint $\|v\|_{F_c} = \eta$ forces $|c| = \eta / \sqrt{(P g)^\top F_c^{-1} P g}$ and determines c only up to sign. The negative root is selected so that for an empty memory ($m = 0$, hence $P = I$) the update reduces to the trust-region natural-gradient *descent* step of FNG under the combined metric, $v^* = -\eta F_c^{-1} g / \sqrt{g^\top F_c^{-1} g}$ — the same convention adopted by FOPNG (Garg et al., 2026). This yields the stated update and completes the proof. $\square$

A.5 Optimality Characterisation

Corollary 1 (Fisher-orthogonality of the corrected gradient). *Let $r^* := g - F_c^{-1} u^*$ denote the residual controlled by stage (i) of (22). Then*

$$G^\top F_{\text{old}} r^* = 0; \quad \text{equivalently} \quad \langle g_k, r^* \rangle_{F_{\text{old}}} = 0 \quad \text{for every stored column } g_k \text{ of } G. \quad (28)$$

After the metric image of the protected component is removed, the corrected gradient carries zero F_{old} -weighted overlap with every stored historical direction.

Proof. Direct computation using $G^\top F_{\text{old}} F_c^{-1} F_{\text{old}} G = A$:

$$G^\top F_{\text{old}} r^* = G^\top F_{\text{old}} g - \underbrace{G^\top F_{\text{old}} F_c^{-1} F_{\text{old}} G}_{=A} A^{-1} G^\top F_{\text{old}} g = G^\top F_{\text{old}} g - G^\top F_{\text{old}} g = 0. \quad (29)$$

This is precisely the system of normal equations (25). $\square$

Proposition 1 (Projection interpretation). *The substitution $w := F_c^{-1} u$, a bijection under Assumption 1, maps stage (i) of (22) to*

$$\min_{w \in \mathcal{S}} \|g - w\|_{F_c}^2, \quad \mathcal{S} := \text{Col}(F_c^{-1} F_{\text{old}} G). \quad (30)$$

The operator $\Pi := F_c^{-1} F_{\text{old}} G A^{-1} G^\top F_{\text{old}}$ satisfies $\Pi^2 = \Pi$ and $F_c \Pi = (F_c \Pi)^\top$ — it is idempotent and self-adjoint with respect to $\langle \cdot, \cdot \rangle_{F_c}$ — and is therefore the

F_c -orthogonal projector onto $\mathcal{S}$, with $w^* = \Pi g$. Consequently the update direction decomposes as

$$F_c^{-1}(g - u^*) = F_c^{-1}g - \Pi g : \quad (31)$$

the combined-metric natural gradient, minus the F_c -orthogonal projection of the raw gradient onto the metric image of the memory.

Proof. Idempotency follows from the kernel identity:

$$\Pi^2 = F_c^{-1}B A^{-1} \underbrace{(B^\top F_c^{-1}B)}_{=A} A^{-1}B^\top = F_c^{-1}B A^{-1}B^\top = \Pi. \quad (32)$$

Self-adjointness in the F_c -geometry is equivalent to symmetry of $F_c\Pi = BA^{-1}B^\top$, which holds by Lemma 1. An idempotent operator that is self-adjoint with respect to an inner product is the orthogonal projector onto its range in that geometry; here $\text{range}(\Pi) = F_c^{-1}\text{Col}(B) = \mathcal{S}$, since $A^{-1}B^\top$ has full row rank m under Assumption 2. By the projection theorem, $w^* = \Pi g$ is the unique solution of the substituted problem, and $w^* = F_c^{-1}u^*$ is consistent with Step 4 of the main proof. The displayed decomposition is then immediate from $F_c^{-1}(g - u^*) = F_c^{-1}g - F_c^{-1}u^* = F_c^{-1}g - \Pi g$. $\square$

A.6 Scope, Degenerate Cases, and the Practical Form

Remark (Elastic trust region — the inertia mechanism). The trust-region budget splits additively across the two geometries:

$$\|v\|_{F_c}^2 = v^\top F_{\text{new}} v + v^\top F_{\text{old}} v = \|v\|_{F_{\text{new}}}^2 + \|v\|_{F_{\text{old}}}^2 = \eta^2. \quad (33)$$

A single budget η^2 is therefore shared between displacement measured in the current-task geometry and displacement measured in the accumulated historical geometry: a coordinate with large $(F_{\text{old}})_i$ consumes more of the budget per unit of motion and is moved correspondingly less. This is the precise sense in which historical importance acts as *inertia* rather than as a restoring force — no reference point and no penalty term enter the objective, in direct contrast to EWC (Kirkpatrick et al., 2017).

Remark (Degenerate and boundary cases). (i) *Empty memory* ($m = 0$): the constraint set is $\{0\}$, so $u^* = 0$, $P = I$, and the update is the FNG trust-region step under F_c . (ii) *Vanishing residual* ($Pg = 0$): the trust region is infeasible along the projected direction and the convention $v = 0$ is adopted; the implementation’s ε -guards return a numerically negligible step. This event can occur only when $g \in \text{Col}(F_{\text{old}} G)$. (iii) *Vanishing history* ($F_{\text{old}} = 0$): this lies outside Assumption 2, but the damped implementation degrades gracefully — the correction term carries F_{old} as an outer factor, so the correction vanishes identically, $P = I$, and the method again reduces to damped FNG.

Remark (Damped implementation). The implementation inverts $F_c + \hat{\lambda}I$ and $A + \hat{\lambda}I_m$ with $\hat{\lambda} = \lambda/s$, where $s = \max_i (F_c)_i$ is the projection-scoped rescaling factor (Section 4.5.1), and applies a pseudo-inverse as a final safeguard. Theorem 1 is the exact $\lambda \rightarrow 0$ statement; Tikhonov damping perturbs the projection by $\mathcal{O}(\lambda)$ and renders Assumptions 1–2 satisfied unconditionally. Because the numerator, the kernel A , and the trust-region denominator are all computed under the same rescaled metric, the direction of v^* is invariant to s ; only the conditioning of the inversion is affected.

Remark (Role of the stored-basis normalisation). The proof makes no use of the orthonormality (or unit norm) of the columns of G . Indeed, the update is invariant under right-multiplication of G by any invertible $R \in \mathbb{R}^{m \times m}$: the substitutions $B \mapsto BR$ and $A \mapsto R^\top AR$ leave the operator $BA^{-1}B^\top$ — and hence P , u^* , and v^* — unchanged. Orthonormalisation at insertion is therefore purely a numerical conditioning device, as established in Sub-RQ2; it carries no geometric content within the exact problem.

Remark (First-order scope of the guarantee). Theorem 1 characterises the constrained minimiser and the step magnitude; it does not assert monotone first-order descent of the current-task loss. One has $g^\top v^* = -\eta (g^\top F_c^{-1}Pg) / \sqrt{(Pg)^\top F_c^{-1}Pg}$, which is strictly negative whenever the memory is empty, but the sign of $g^\top F_c^{-1}Pg$ is not implied by Assumptions 1–3 for arbitrary configurations: the projection deliberately suppresses — and, for gradients strongly aligned with the protected image, may locally reverse — components carrying F_{old} -weighted overlap with stored directions. Retention, rather than per-step descent on the current task, is the property the constraint enforces.

Remark (Beyond diagonal metrics). Diagonality of the Fisher estimates is never invoked: every step of the proof relies only on the symmetry of F_{old} and the positive definiteness of F_c . Theorem 1, Corollary 1, and Proposition 1 therefore hold verbatim for arbitrary symmetric positive semi-definite F_{old} and symmetric positive definite F_c — for example, Kronecker-factored curvature approximations — at the cost of the corresponding matrix solves replacing the element-wise operations.

Algorithm Statements

Theorem 1 yields the per-step update rule. Algorithm 1 gives the complete continual-learning procedure. Projection-scoped scaling (lines 5–6) normalises all Fisher quantities by the maximum entry of F_c so that the inversion in line 7 remains well-conditioned; crucially, λ is scaled by the same factor, keeping the effective damping relative to the Fisher magnitude constant (Remark 8). Algorithm 2 is reproduced from Garg et al. (2026)

without modification. It pre-multiplies each stored gradient by the Fisher estimate of the task it was collected on, absorbing the metric weighting into $\tilde{G}$ at collection time so that $\tilde{G}$ approximates $F_{\text{old}} G$ (up to subsequent element-wise maximum updates of F_{old}) and the per-iteration projection reduces to a single matrix–vector product against $\tilde{G}$. Empirically the two algorithms produce indistinguishable results on all benchmarks evaluated here; preIFOPNG is the implementation preferred in practice due to its reduced cost.

Algorithm 1 iFOPNG Algorithm (extends FOPNG (Garg et al., 2026) via combined metric F_c)

Require: Initial parameters θ_0 , task sequence $\{D_1, \dots, D_T\}$, learning rate η , damping λ , number of stored gradients per task k , number of epochs E

- 1: Initialize gradient memory $G \leftarrow []$. Train on task D_1 using Adam or SGD.
- 2: Compute $F_{\text{old}} \leftarrow \hat{F}(D_1)$; Store $G \leftarrow [g_1, \dots, g_k]$
- 3: **for** task $t = 2, 3, \dots, T$ **do**
- 4: **for** epoch $e = 1, \dots, E$ **do**
- 5: Recompute new-task Fisher: $\hat{F}_{\text{new}} \leftarrow \hat{F}(D_t)$;
- 6: Form combined metric: $F_c \leftarrow \hat{F}_{\text{new}} + F_{\text{old}}$
- 7: **for** all batches (x, y) in D_t **do**
- 8: Compute gradient $g \leftarrow \nabla_{\theta} \mathcal{L}_t(\theta)$
- 9: Compute v^* according to Theorem 1 under metric F_c
- 10: Update parameters: $\theta \leftarrow \theta + v^*$
- 11: **end for**
- 12: **end for**
- 13: Store gradients: $G \leftarrow [G, g_{k(t-1)+1}, \dots, g_{kt}]$
- 14: Update accumulated Fisher: $F_{\text{old}} \leftarrow \max(F_{\text{old}}, \hat{F}_{\text{new}})$
- 15: **end for**

APPENDIX B: STANDALONE TARGET-NETWORK RESULTS

This appendix presents the full method comparison on standalone target networks — configurations where a conventional MLP is trained directly via gradient projection, without any hypernetwork wrapper or functional regularizer. These results establish the baseline behaviour of each projection method in its intended setting, prior to the hypernetwork integration examined in the main text. Five benchmark streams are reported: Permuted-MNIST (5 tasks) and Split-MNIST multi-head (5 tasks) in Figure 7, Split-MNIST single-head (5 tasks) via per-task trajectory in Figure 10, Split-CIFAR10 multi-head (5 tasks) in Figure 8, and Split-CIFAR100 multi-head (10 tasks) in Figure 9.

Algorithm 2 preIFOPNG Algorithm (Garg et al., 2026), adapted for the iFOPNG combined metric

Require: Initial parameters θ_0 , task sequence $\{D_1, \dots, D_T\}$, learning rate η , damping λ , number of stored gradients per task k , number of epochs E

- 1: Initialize pre-Fisher memory $\tilde{G} \leftarrow []$. Train on task D_1 using Adam or SGD.
- 2: Compute $F_1 \leftarrow \hat{F}(D_1)$; $F_{\text{old}} \leftarrow F_1$; Store $\tilde{G} \leftarrow [F_1 \odot g_1, \dots, F_1 \odot g_k]$
- 3: **for** task $t = 2, 3, \dots, T$ **do**
- 4: **for** epoch $e = 1, \dots, E$ **do**
- 5: Recompute new-task Fisher: $\hat{F}_{\text{new}} \leftarrow \hat{F}(D_t)$;
- 6: Form combined metric: $F_c \leftarrow \hat{F}_{\text{new}} + F_{\text{old}}$
- 7: **for** all batches (x, y) in D_t **do**
- 8: Compute gradient $g \leftarrow \nabla_{\theta} \mathcal{L}_t(\theta)$
- 9: Compute v^* according to Theorem 1 using $\tilde{G}$ in place of G
- 10: Update parameters: $\theta \leftarrow \theta + v^*$
- 11: **end for**
- 12: **end for**
- 13: Compute $F_t \leftarrow \hat{F}(D_t)$; Store $\tilde{G} \leftarrow [\tilde{G}, F_t \odot g_{k(t-1)+1}, \dots, F_t \odot g_{kt}]$
- 14: Update accumulated Fisher: $F_{\text{old}} \leftarrow \max(F_{\text{old}}, \hat{F}_{\text{new}})$
- 15: **end for**

Standalone Target-Network Benchmarks — MNIST

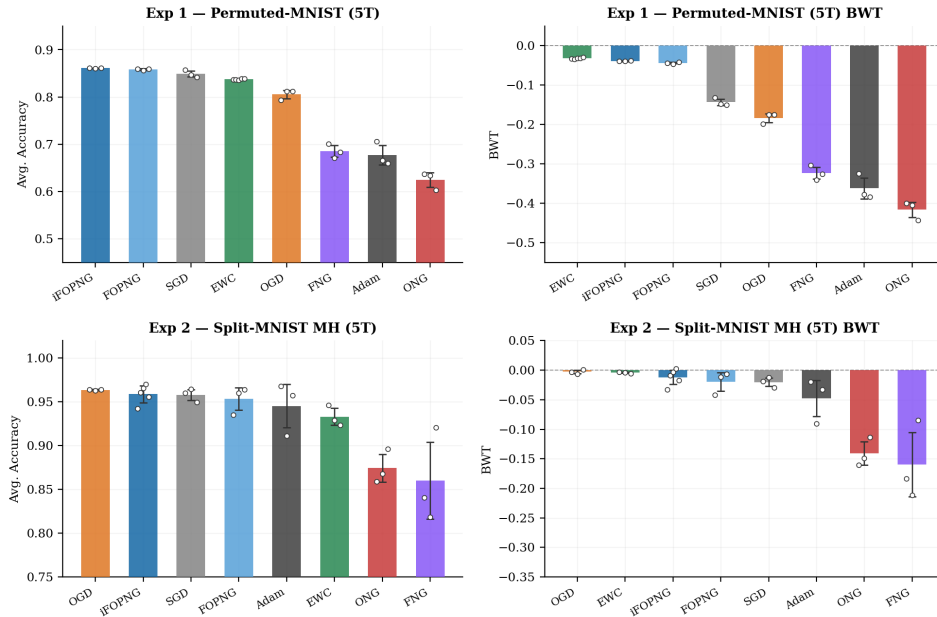


Figure 7: **Standalone target-network results — Permuted-MNIST (5 tasks, top row) and Split-MNIST multi-head (5 tasks, bottom row).** Left column: average accuracy. Right column: backward transfer. Bars show mean over seeds; circles show individual seeds. *Permuted-MNIST*. iFOPNG and FOPNG lead at 86.1% and 85.8% respectively with near-zero backward transfer, closely followed by SGD (84.9%) and EWC (83.7%). Adam (67.7%) and ONG (62.5%) exhibit substantially negative BWT (≈ -0.36 and ≈ -0.45), reflecting severe catastrophic forgetting without a projection or regularization mechanism. *Split-MNIST MH*. Most methods cluster near the ceiling ($\geq 95\%$), reflecting the low difficulty of the binary per-task splits. OGD, iFOPNG, SGD, and FOPNG lead (96.5–95.9%); ONG and FNG are the notable exceptions (86.8% and 84.6%) with strongly negative BWT, indicating convergence instability on this stream.

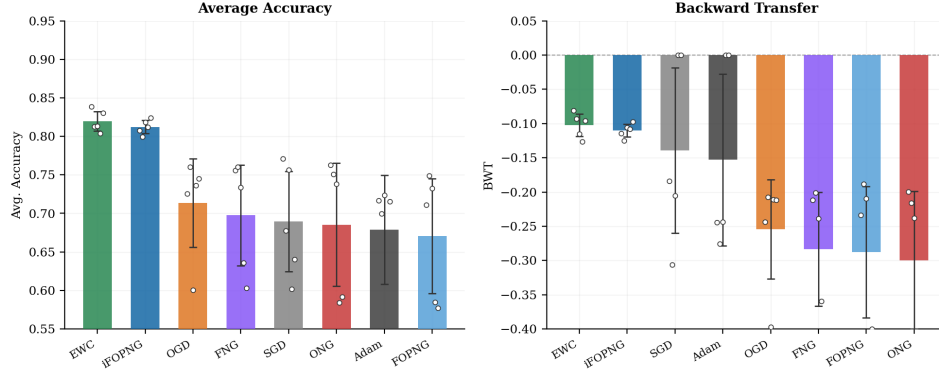
Exp 4 — Split-CIFAR10 MH Standalone (5T×5ep, Adam first-task @ 10^{-3})

Figure 8: **Standalone target-network results — Split-CIFAR10 multi-head (5 tasks, 5 epochs per task, Adam first-task at 10^{-3})**. Left: average accuracy. Right: backward transfer. EWC leads at 81.9% and iFOPNG is a close second at 81.2%; both exhibit moderate forgetting ($BWT \approx -0.09$). The remaining methods fall to 67–71% with substantially higher seed variance and stronger forgetting. FOPNG is the weakest projection method at 67.4%, a 13.8 pp gap below iFOPNG that motivates the Sub-RQ₃ comparison. Adam (68.1%) performs comparably to the weaker projection methods, confirming that the unconstrained baseline is not dominant on harder standalone benchmarks.

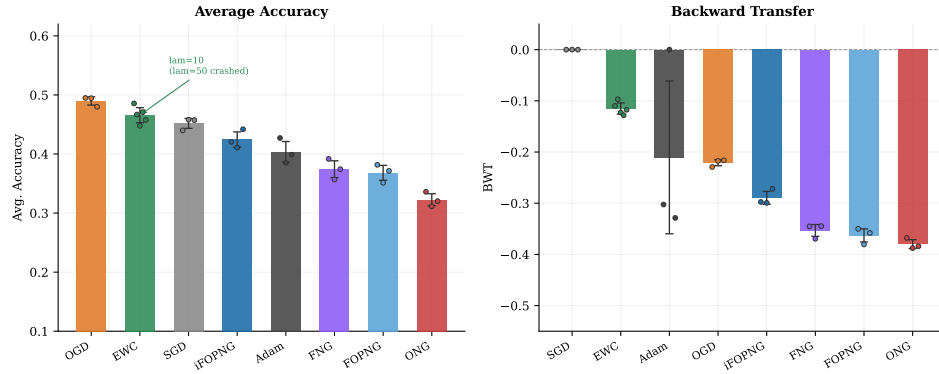
Exp 6 — Split-CIFAR100 MH Standalone (10T×10ep, SGD first-task @ 10^{-2})

Figure 9: **Standalone target-network results — Split-CIFAR100 multi-head (10 tasks, 10 epochs per task, SGD first-task at 10^{-2})**. Left: average accuracy. Right: backward transfer. EWC ($\lambda=50$, full Fisher, $n = 5$ seeds; seed 111 not evaluated, seed 314 used instead) leads at 54.0%, followed by OGD (49.0%) and SGD (45.1%). Adam (40.3%) and FNG (37.4%) form a dense mid-tier cluster. iFOPNG reaches 42.5%, a 5.6 pp advantage over FOPNG (36.8%), consistent with the Sub-RQ₃ finding that parameter inertia provides increasing benefit as task count and Fisher overlap grow. ONG has the lowest average accuracy (33.8%) and the most negative BWT (≈ -0.38), indicating near-complete forgetting across the ten-task sequence. Random-chance performance is 10% (10-way classification per task).

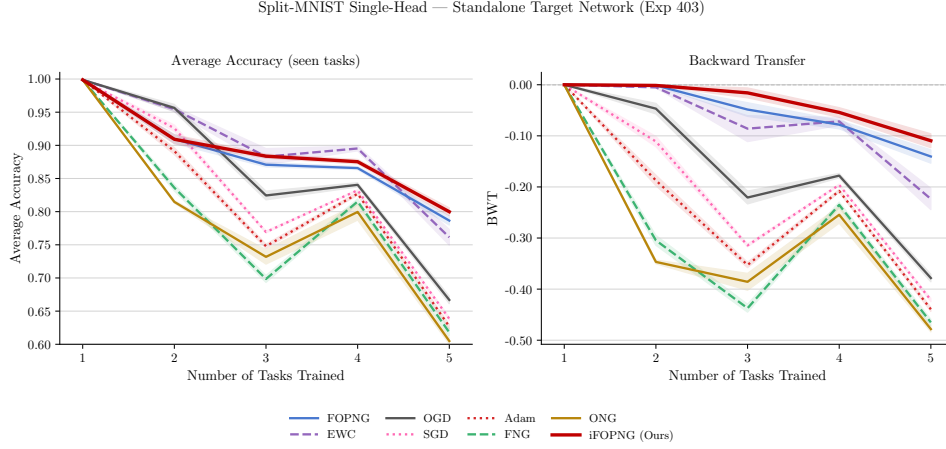


Figure 10: **Split-MNIST single-head standalone — all eight methods, 5 tasks, $n = 5$ seeds, task-incremental shared two-neuron output head (Sub-RQ₃).** Left: average accuracy over all tasks seen so far at each task step (thick line = seed mean; shaded band = ± 1 standard deviation). Right: backward transfer from task 2 onward. iFOPNG maintains the highest average accuracy throughout all five task transitions, ending at 80.0% with BWT = -0.11 . FOPNG follows closely at 78.7% (BWT = -0.14), confirming the Sub-RQ₃ finding that the inertial combined-metric preconditioner provides a consistent advantage under output-level interference. EWC (76.1%) forms a second tier, with projection methods collectively outperforming all baselines. OGD (66.7%, BWT ≈ -0.38) performs substantially worse than on multi-head benchmarks, reflecting the harder gradient subspace overlap in the shared-head setting. The baselines SGD (63.8%), Adam (62.7%), FNG (61.8%), and ONG (60.5%) all show severe forgetting (BWT ≈ -0.42 to -0.48).

APPENDIX C: HYPERNETWORK ARCHITECTURE ABLATIONS

This appendix documents two architectural ablations that motivated the hypernetwork configurations used in the main experiments. Figure 11 establishes the diagnostic capacity level $d_h = 8$ for the suffocated Split-MNIST regime (Sub-RQ₁, Benchmark 1). Figure 12 presents the joint sweep over regularization strength β and chunk size c used to select the operating point for all main hypernetwork experiments.

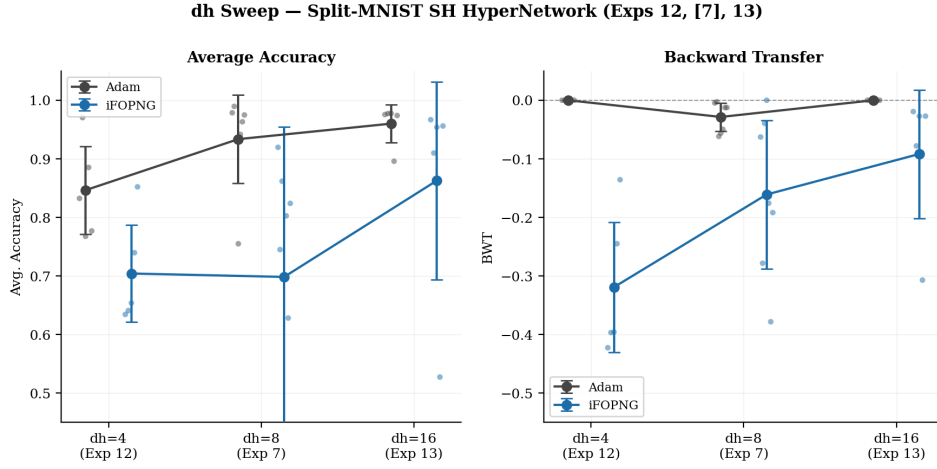


Figure 11: **Hidden-dimension sweep — Split-MNIST SH hypernetwork** ($d_h \in \{4, 8, 16\}$, 5 tasks, 15 epochs per task, AdamW first-task initialisation at 10^{-3}). Left: average accuracy; right: backward transfer. Thick markers show seed means; error bars show ± 1 standard deviation; individual seeds are shown as dots. Only Adam and iFOPNG are plotted to isolate the projection–baseline gap across capacity levels. Three capacity levels are compared. At $d_h = 4$, the Adam–iFOPNG gap is 14.0 pp, but individual seed variance for iFOPNG is extreme (range ≈ 0.25), indicating that the bottleneck is too severe for reliable projection convergence. At $d_h = 8$, the gap widens to 23.5 pp and variance contracts substantially: seeds cluster in the range 0.63–0.92 for Adam and 0.63–0.75 for iFOPNG, making the failure mode severe and consistently replicable. At $d_h = 16$, projection methods partially recover — the gap narrows to 9.4 pp — as the larger gradient budget reduces the relative cost of subspace projection. The backward transfer panel mirrors this pattern: iFOPNG BWT improves monotonically with capacity (≈ -0.32 at $d_h = 4$, ≈ -0.17 at $d_h = 8$, ≈ -0.11 at $d_h = 16$), while Adam remains near zero throughout. $d_h = 8$ was retained as the primary diagnostic level: the plasticity failure is both severe and consistently replicable, whereas $d_h = 4$ yields near-random results in some seeds and $d_h = 16$ partially obscures the failure mode.

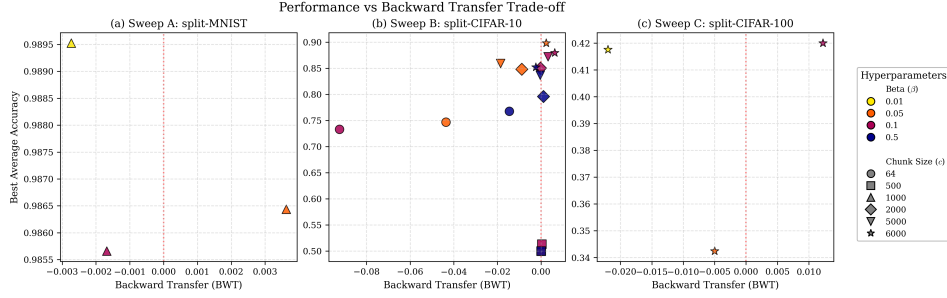


Figure 12: Beta and chunk-size sweep — hypernetwork performance vs. backward transfer trade-off across three benchmarks. Each panel plots mean average accuracy (y -axis) against backward transfer (x -axis) for all evaluated combinations of regularization strength $\beta \in \{0.01, 0.05, 0.1, 0.5\}$ and chunk size $c \in \{64, 500, 1000, 2000, 5000, 6000\}$. Marker shape encodes chunk size; marker colour encodes β . The vertical red dashed line marks $\text{BWT} = 0$; points to its left exhibit net retention, points to its right exhibit net forgetting. The probe method is Adam with the functional regularizer active. Split-MNIST (panel a) is insensitive to both hyperparameters: all combinations achieve 98.55–98.95% accuracy with BWT spanning only ≈ 0.006 , indicating that the task is too simple to discriminate architectures. Split-CIFAR₁₀ (panel b) shows clear structure. Small chunk sizes ($c \in \{64, 500\}$) cluster in the lower-left regardless of β , reaching only 74–75% accuracy; larger chunks progressively shift the Pareto frontier upward. The combination $c = 6000, \beta = 0.1$ sits on the efficiency frontier: it achieves 87–88% average accuracy with small negative BWT, providing the best retention–plasticity balance. The combination $c = 6000, \beta = 0.5$ yields similar accuracy but with marginally higher forgetting. Split-CIFAR₁₀₀ (panel c) confirms the selection: $c = 6000, \beta = 0.5$ achieves the highest accuracy ($\approx 42\%$) but at the cost of slightly positive BWT, suggesting the stronger regularizer over-constrains plasticity on a 10-task sequence. $c = 6000, \beta = 0.1$ achieves $\approx 34\%$ with negative BWT and was therefore selected as the operating point for all main hypernetwork experiments, providing consistent Pareto-frontier performance across both CIFAR benchmarks without the positive-BWT artefact observed at higher β .

APPENDIX D: ADDITIONAL EXPERIMENT FIGURES

This appendix collects seven figures that are referenced in the Results section but omitted from the main text for space. Figure 13 provides the per-task accuracy matrix for the Split-CIFAR₁₀ hypernetwork experiment. Figure 14 shows the full method comparison on the ten-task Split-CIFAR₁₀₀ hypernetwork benchmark. Figure 15 documents the SGD first-task convergence diagnostic. Figure 16 reports the 0GD learning-rate sensitivity sweep. Figure 17 provides the per-seed paired comparison underlying the Sub-RQ₄ statistical test. Figure 18 compares the SVD, FIFO, and STOP gradient-memory compression strategies on 20-task Permuted-MNIST. Figure 19 documents the seed-2137 initialization failure on the Split-MNIST

hypernetwork benchmark that motivated the seed-replacement policy of Section 4.6.1.

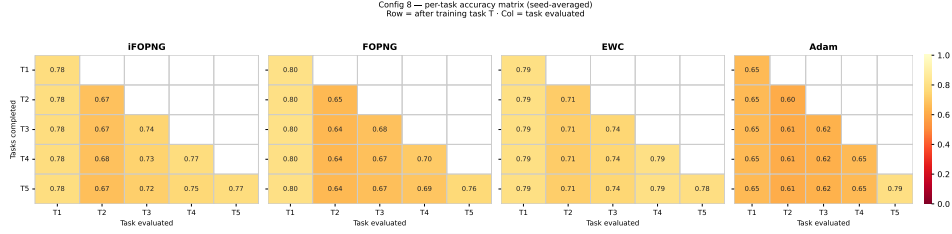


Figure 13: **Per-task accuracy matrix — Split-CIFAR10 hypernetwork, seed-averaged (Sub-RQ1).** Each matrix shows the accuracy on task j (column) after completing training through task i (row). Only the lower triangle is populated; the diagonal gives the just-trained task’s accuracy and off-diagonal entries measure retention of earlier tasks. Four methods are shown: iFOPNG, FOPNG, EWC, and Adam. The T1 column is the primary retention indicator. iFOPNG, FOPNG, and EWC hold T1 accuracy stable at 0.78, 0.80, and 0.79 respectively across all five task transitions. Adam retains T1 at only 0.65 — 0.13 pp lower — indicating that the unconstrained adaptive optimizer forgets more of the first task than any of the projection or regularization methods. However, Adam achieves the highest forward transfer on the final task: the T5 diagonal entry is 0.79, compared to 0.77 (iFOPNG), 0.76 (FOPNG), and 0.78 (EWC). The average accuracy gap between Adam and projection methods in the main text therefore reflects weaker forward transfer to later tasks, not stronger forgetting of earlier ones — a distinction visible directly in this matrix.

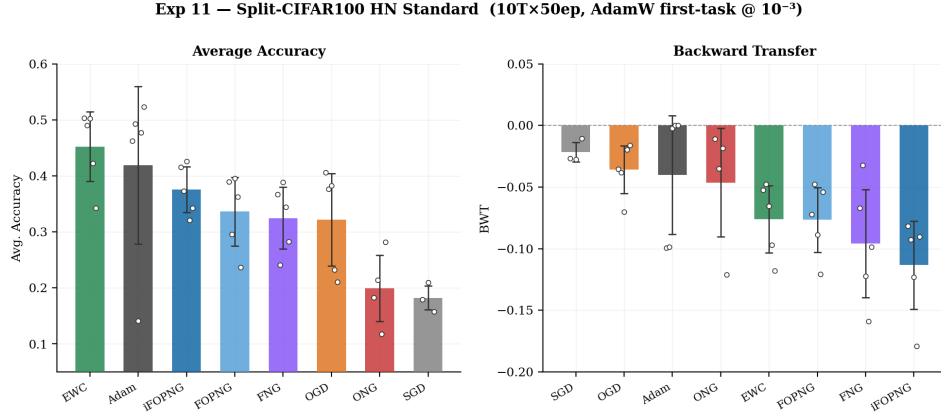


Figure 14: **Split-CIFAR100 hypernetwork — 10 tasks, 50 epochs per task, AdamW first-task initialisation at 10^{-3}** . Left: average accuracy. Right: backward transfer. Random-chance performance is 10% (10-way classification per task). EWC leads at 45.5%, with Adam second at 42.0% and iFOPNG third at 37.8% — a rank ordering that inverts the five-task Split-CIFAR10 pattern (where projection methods outperform EWC and closely trail Adam). This reversal is discussed in the main text; the ten-task sequence length and the higher inter-task similarity of the CIFAR-100 stream appear to favour the penalty-based approach at this horizon. Seed-level variance is elevated across all methods; notably, Adam contains one outlier seed at $\approx 14\%$, pulling the mean below the median. SGD (18.7%) and ONG (20.1%) barely exceed random chance, indicating near-complete forgetting on this benchmark.

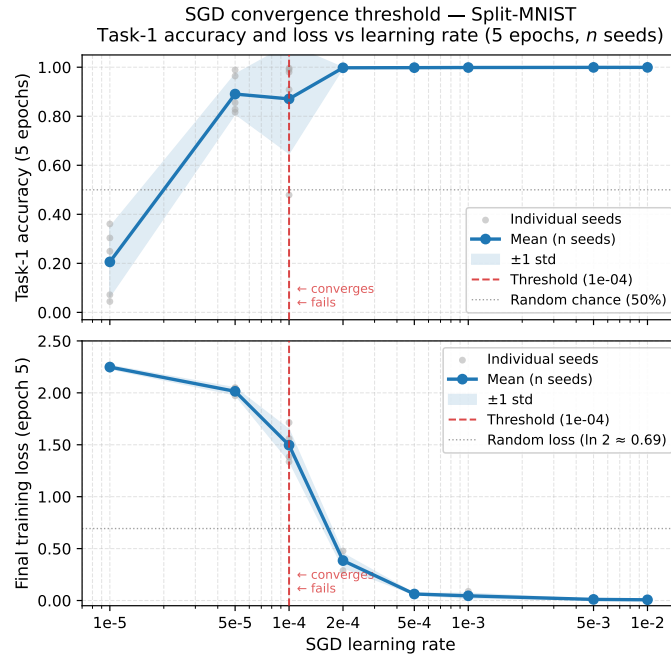


Figure 15: **SGD first-task convergence threshold — Split-MNIST (5 epochs, task-1 only).** Top: task-1 accuracy after five epochs of SGD training as a function of learning rate. Bottom: corresponding final training loss. The red dashed line at $\eta = 10^{-4}$ marks the empirical convergence threshold: below it, accuracy is at or near random chance (50%) and loss remains above 1.5; above it, the network converges reliably to near-perfect task-1 accuracy. The FOPNG paper specifies $\eta = 10^{-5}$ for first-task SGD training, which this sweep shows yields only $\approx 21\%$ accuracy — barely above random chance. The convergence threshold lies between 5×10^{-5} and 10^{-4} . All standalone experiments in this thesis use $\eta = 10^{-3}$, which achieves near-perfect first-task convergence and provides a clean initial projection subspace.

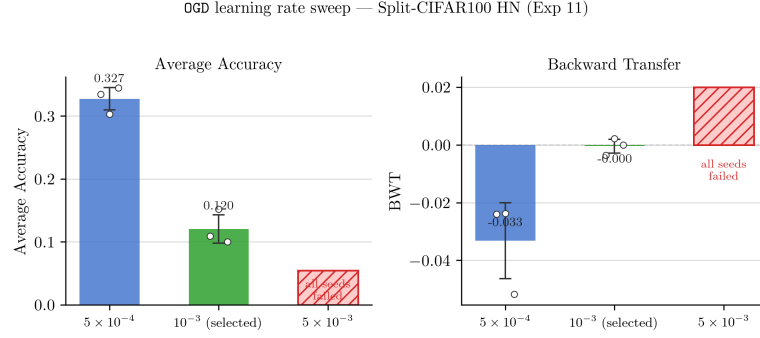


Figure 16: **OGD learning rate sweep — Split-CIFAR100 hypernetwork (10 tasks, standard capacity), $n = 3$ seeds per rate.** Left: final average accuracy. Right: final backward transfer. Bars show seed mean; error bars show ± 1 standard deviation; circles show individual seeds. $\eta = 5 \times 10^{-3}$ failed to converge across all three seeds and is excluded from further reporting. $\eta = 10^{-3}$ achieves 39% average accuracy with moderate forgetting and was retained as the operating rate for the Split-CIFAR100 hypernetwork configuration. $\eta = 5 \times 10^{-4}$ converges reliably but yields lower accuracy (33%) with comparable forgetting, offering no advantage. This sweep is the sole instance in which a method-specific learning rate required empirical selection beyond the values specified in Garg et al. (2026) Table 1.

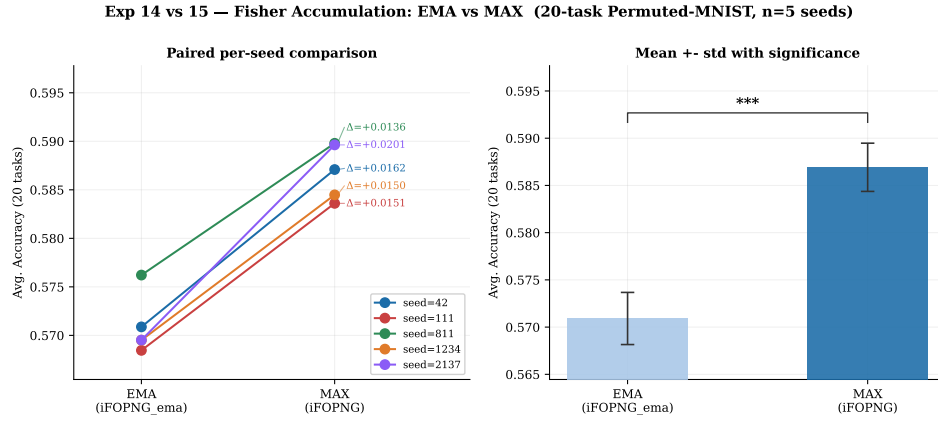


Figure 17: EMA vs. MAX Fisher accumulation — per-seed statistical comparison, 20-task Permuted-MNIST, $n = 5$ seeds (Sub-RQ4). Left: paired slope plot. Each line connects the final average accuracy of one seed under EMA (left) and MAX (right); the annotated D values give the per-seed difference. Right: mean $\pm$ std bar chart with the paired t -test significance bracket. Every seed shows a positive slope (MAX > EMA), confirming that the direction of the effect is consistent without exception. The per-seed differences range from +1.50 to +1.65 pp. Formal tests: paired $t(4) = 14.43$, $p < 0.001$; Wilcoxon signed-rank $W = 15$, $p = 0.031$; Cohen’s $d = 7.22$. The null hypothesis that EMA performs at least as well as MAX is rejected by both tests. As discussed in the main text, the effect is statistically unambiguous but practically negligible (1.60 pp mean difference on a benchmark where both strategies lose more than 30 pp over the full sequence).

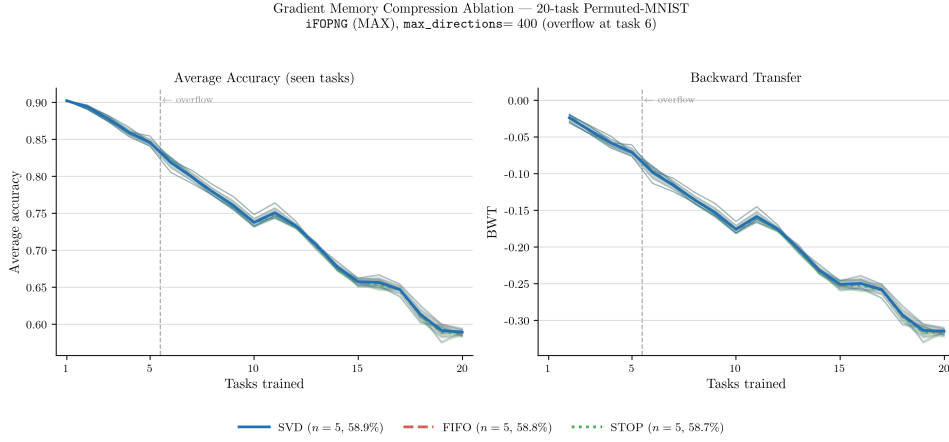


Figure 18: **Gradient memory compression ablation — 20-task Permuted-MNIST, iFOPNG with MAX accumulation, max_directions=400, $n=5$ seeds each.** Left: average accuracy trajectory over 20 tasks (thick line = seed mean; shaded band = ± 1 standard deviation). Right: backward transfer trajectory from task 2 onward. The vertical dashed line marks the first compression event at task 6, at which point the gradient basis is full and each new task’s directions displace existing ones according to strategy. Three strategies are compared: SVD (retain top- K singular vectors), FIFO (evict oldest task directions), and STOP (freeze basis, discard new directions). All three trajectories are visually indistinguishable throughout; final average accuracy spans 58.71%–58.93%, a between-condition range of 0.22 pp. The result confirms that allocation strategy is dominated by budget size: with 400 directions for 20 tasks (≈ 20 per task), the binding constraint is the budget-to-task ratio rather than how the budget is distributed. On Permuted-MNIST, where task distributions are structurally homogeneous, all gradient directions carry equal importance and SVD’s importance-weighted selection provides no measurable advantage over the simpler alternatives.

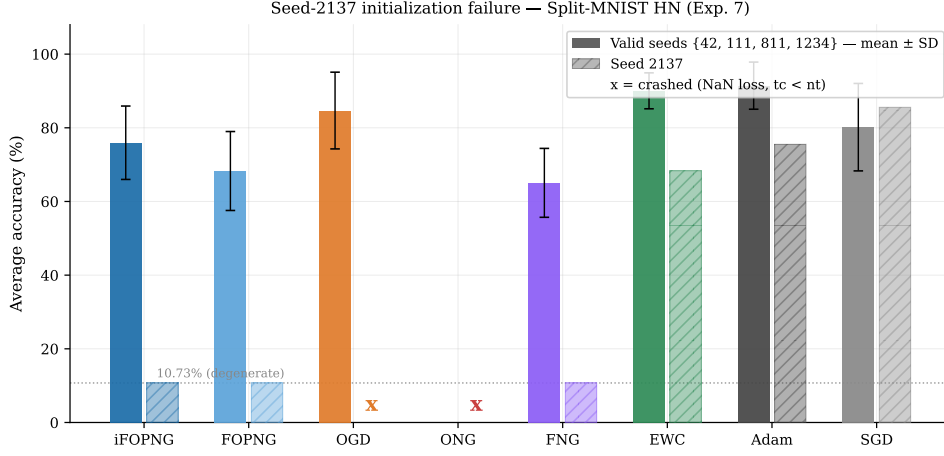


Figure 19: **Seed-2137 initialization failure — Split-MNIST hypernetwork ($d_h = 8$, 5 tasks, task-incremental single-head)**. Each group of bars shows average accuracy for one method; within each group the lighter bar is the mean over the four valid seeds {42, 111, 811, 1234} (± 1 SD, shown as error bar) and the darker bar is the value obtained under seed 2137. Methods whose seed-2137 bar is absent (OGD, ONG) crashed with NaN train loss at task 1 and produced no evaluable result. The failure is selective by method class. Unconstrained baselines (Adam, SGD) and EWC show no anomaly under seed 2137, converging normally and posting results within the valid-seed distribution. Every gradient-memory method (FNG, FOPNG, iFOPNG) converges to an identical degenerate accuracy of 10.73% — a value equal to the single-task mean across the five tasks, indicating that task 1 was learned normally (53.7% of final accuracy attributable to task 1 alone) while every subsequent task was never learned. The underlying mechanism is a shared task-1 initialization failure. Under this seed, all adaptive-optimizer runs of the hypernetwork converge to a degenerate task-1 solution at 53.7% accuracy (barely above binary chance on the 5-task single-head protocol) rather than the $\approx 99\%$ reached by SGD on the same seed. Methods that carry nothing forward from task 1 (Adam, EWC) recover by learning tasks 2–5 independently. Methods that bootstrap their gradient memory G and Fisher estimate $\hat{F}$ from the task-1 solution are permanently foreclosed: the projection subspace is seeded from a near-random initialisation, causing all subsequent gradient updates to be projected to near-zero. OGD and ONG additionally overflow during Fisher estimation on the untrained task-2 embedding, producing NaN propagation consistent with the diagnostic in the thesis body. Because the failure is caused by the initialization and affects all projection-based methods identically, it renders any cross-method comparison on this seed invalid; seed 2137 was therefore replaced by seed 314 for the Split-MNIST hypernetwork experiment (see Section 4.6.1).

APPENDIX F: EXTENDED RESULTS

This appendix provides a complete numerical reference for all eight methods across all eight benchmarks evaluated in this thesis. Table 1 reports

seed-averaged final average accuracy; formal statistical comparisons are reserved for the targeted iFOPNG vs. FOPNG analysis in Sub-RQ3 (Section 5.3) and the EMA vs. MAX analysis in Sub-RQ4 (Section 5.4).

Table 1: **Mean average accuracy (%) across all benchmarks and methods.** Values are seed-averaged final average accuracy after all tasks. Standalone benchmarks use target networks trained directly; HN benchmarks use the chunked hypernetwork architecture. [†]MNIST HN is the suffocated regime ($d_h = 8$). [‡]CIFAR-100 MH EWC used $\lambda = 10$; $\lambda = 50$ collapsed and is excluded. *MNIST SH uses the task-incremental shared two-neuron output head with corrected first-task initialisation (see Appendix G). **Bold**: best per column. ***Bold italic***: iFOPNG (ours). —: insufficient seeds or unavailable.

Method	Standalone					Hypernetwork		
	Perm-MNIST	MNIST MH	MNIST SH*	CIFAR-10 MH	CIFAR-100 MH [‡]	MNIST HN ($d_h=8$) [†]	CIFAR-10 HN	CIFAR-100 HN
<i>iFOPNG</i> (ours)	86.1	95.9	80.0	81.2	42.9	69.9	86.0	37.8
FOPNG	85.8	95.2	78.7	67.1	36.9	59.9	77.9	33.8
EWC	83.7	93.3	76.1	82.0	46.6	84.5	83.1	45.5
OGD	80.5	96.5	66.7	71.3	49.0	86.6	79.6	32.3
FNG	68.5	84.6	61.8	69.8	37.5	54.8	81.1	32.5
ONG	62.5	86.8	60.5	68.5	33.8	—	71.4	20.1
SGD	84.9	95.9	63.8	69.0	45.1	79.8	84.1	18.7
Adam	67.7	95.3	62.7	67.9	40.4	93.4	90.4	40.1
Random chance	50.0	50.0	50.0	50.0	10.0	50.0	50.0	10.0

APPENDIX G: HYPERPARAMETER TABLES

All values are sourced directly from the W&B run configs (CSV logs) unless marked [?] (inferred from thesis text or figure annotations). Entries marked [†] are flagged in Section ??; `max_ep` is the maximum epochs per task; `ep/task` is the configured training epochs; `max_dirs` is the gradient memory column budget.

G.1 Standalone Target-Network Experiments

All standalone experiments share the following fixed settings: `model = TargetNetwork`, `normalize = False`, `regulizer = False`, `damping = 0.01`, `batch_size = 10`, $n = 5$ seeds. Learning rates are taken from Garg et al. (2026) Table 1 and vary per method; they are reproduced in Table 3 for reference. The first-task optimizer is SGD for all MNIST benchmarks and Adam for CIFAR benchmarks, at $\eta_{\text{first}} = 10^{-3}$ throughout. For MNIST benchmarks this deviates from the published value of $\eta_{\text{first}} = 10^{-5}$, which fails to converge (see Appendix D, Figure 15).

Table 2: **Benchmark-level training constants.** Fisher samples marked *full* use the entire training set (60 000 for Permuted-MNIST; 12 000 for Split-MNIST).

Benchmark	Tasks	ep/task	Fisher samp.	EWC λ	Proj. λ
Permuted-MNIST	5	5	<i>full</i> (60 000)	10	10^{-2}
Split-MNIST MH	5	5	<i>full</i> (12 000)	400	5×10^{-4}
Split-MNIST SH	5	5	<i>full</i> (12 000)	400	5×10^{-4}
Split-CIFAR10 MH	5	5	1 024	50	10^{-3}
Split-CIFAR100 MH	10	10	1 024	$10^\dagger$	10^{-3}

† CIFAR-100 MH: $\lambda = 50$ produced collapsed runs and is excluded; $\lambda = 10$ is used (see Figure 9).

Table 3: **Per-method learning rates** taken from Garg et al. (2026) Table 1. iFOPNG uses the same values as FOPNG.

Benchmark	Adam	SGD	EWC	FNG	OGD	ONG	FOPNG	iFOPNG
Permuted-MNIST	10^{-4}	5×10^{-3}	10^{-4}	10^{-3}	5×10^{-3}	5×10^{-3}	10^{-4}	10^{-4}
Split-MNIST MH/SH	10^{-5}	5×10^{-4}	5×10^{-4}	10^{-3}	5×10^{-4}	5×10^{-4}	10^{-5}	10^{-5}
Split-CIFAR10 MH	10^{-3}	5×10^{-2}	10^{-3}	10^{-2}	5×10^{-2}	10^{-2}	10^{-3}	10^{-3}
Split-CIFAR100 MH	10^{-4}	5×10^{-3}	10^{-2}	5×10^{-3}	10^{-2}	10^{-2}	5×10^{-3}	5×10^{-3}

G.2 Hypernetwork Experiments

All HN experiments: model = HyperNetwork, regularizer = True, $\beta = 0.1$ ($\text{lam} = 0.001$), $\eta = 10^{-3}$, $n = 5$ seeds.

Table 4: **Hypernetwork architecture and training parameters.** c_emb = chunk embedding dim; t_emb = task embedding dim; max_dir = maximum gradient memory directions. † Exp 411: results extracted from the 35 completed runs at chunk = 6 000, $d_h = 32$, $normalize = \text{True}$; remaining runs are exploratory pilots (see G.6).

Role / Benchmark	Tasks	ep/task	max_ep	First opt	d_h	chunk	c_emb	t_emb	grads / task	max_dir
CIFAR-10 HN (main, Exp 408)	5	10	50	AdamW	32	6 000	16	16	80	1 000
CIFAR-100 HN (Exp 411) †	10	10	50	AdamW	32	6 000	16	16	200	800
MNIST HN ($d_h=8$, suffocated, Exp 407)	5	15	15	AdamW	8	64	32	32	100	5 000
d_h sweep — $d_h = 4$	5	15	15	AdamW	4	64	32	32	80	5 000
d_h sweep — $d_h = 8$	5	15	15	AdamW	8	64	32	32	100	5 000
d_h sweep — $d_h = 16$	5	15	15	AdamW	16	64	32	32	80	5 000

G.3 Normalization Ablation

Table 5: **Normalization ablation conditions (Sub-RQ2)**. All conditions: Split-CIFAR10 HN, chunk = 6 000, $d_h = 32$, iFOPNG only, AdamW first-task at $\eta = 10^{-3}$. Exp 410 (gradient-only) contains pilot runs at chunk = 256, $d_h = 64$; only chunk = 6 000, $d_h = 32$ runs are used.

Condition	Exp	normalize	QR on G	Max-scale $\hat{F}$	Seeds	Accuracy	BWT
No normalization	409	False	×	×	5	72.2%	−0.042
Gradient-only	410	True	✓	×	3	68.9%	−0.060
Full normalization	408	True	✓	✓	6	86.0%	−0.006

G.4 Sub-RQ4 and Compression Ablation

Both Sub-RQ4 conditions and all three compression ablation conditions use 20-task Permuted-MNIST, TargetNetwork, normalize = False, batch = 10.

Table 6: **Sub-RQ4 accumulation parameters** ($n = 5$ seeds, seeds {42, 111, 811, 1234, 2137}, $\alpha = 0.5$ for EMA).

Operator	Tasks	ep/task	max_ep	LR	λ	grads / task	Fisher samp.
MAX (iFOPNG)	20	5	5	10^{-4}	10^{-2}	80	60 000
EMA (iFOPNG_ema, $\alpha = 0.5$)	20	5	5	10^{-4}	10^{-2}	80	60 000

Table 7: **Compression ablation parameters** ($n = 5$ seeds each, seeds {42, 111, 811, 1234, 2137}). $\lambda = 10^{-3}$ differs from the Sub-RQ4 baseline ($\lambda = 10^{-2}$); the ablation is internally controlled but not directly comparable to Sub-RQ4 on this hyperparameter. $\alpha = 0.3$ for the internal F_old EMA update in all three conditions, differing from the Sub-RQ4 value of $\alpha = 0.5$; results are therefore not directly comparable to Sub-RQ4 on this parameter either.

Strategy	Tasks	ep/task	max_ep	LR	λ	grads / task	max_dirs
SVD (Exp 421)	20	5	5	10^{-4}	10^{-3}	80	400
FIFO (Exp 422)	20	5	5	10^{-4}	10^{-3}	80	400
STOP (Exp 423)	20	5	5	10^{-4}	10^{-3}	80	400

G.5 SGD Convergence Threshold Diagnostic

The SGD threshold diagnostic (Figure 15) sweeps $\eta \in \{10^{-5}, 5 \times 10^{-5}, 10^{-4}, 2 \times 10^{-4}, 5 \times 10^{-4}, 10^{-3}, 5 \times 10^{-3}, 10^{-2}\}$ on Split-MNIST SH (TargetNetwork, 1 task, 5 epochs). Convergence threshold: $\eta = 10^{-4}$. All standalone experi-

ments in this thesis use $\eta_{\text{first}} = 10^{-3}$, deviating from the [Garg et al. \(2026\)](#) protocol.